75

Surgical, Radiographic, and Endoscopic Anatomy of the Retroperitoneum

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Keywords

adrenal; anatomy; kidney; retroperitoneum; ureter

Questions

1. Which of the following structures is NOT in the retroperitoneum?
    a. Kidney
    b. Second portion of the duodenum
    c. Ascending colon
    d. Adrenal
    e. Transverse colon
2. Which muscle's function is most similar to psoas major?
    a. Iliacus
    b. Quadratus lumborum
    c. Transversus abdominis
    d. External oblique
    e. Vastus lateralis
3. Which of the following statements is TRUE?
    a. The intercostal neurovascular bundle travels between the internal intercostal and innermost intercostal muscles.

b. The 10th, 11th, and 12th ribs are floating ribs.
c. Fracture of the lower ribs does not pose a risk to the retroperitoneal structures.
d. Bony landmarks are not useful for planning surgical incisions.
e. The intercostal neurovascular bundle travels within the costal groove on the cephalad aspect of the rib.

4. Which fascial layer is immediately deep to the transversus abdominis muscle?
   a. Lumbodorsal fascia
   b. Lateroconal fascia
   c. Internal oblique fascia
   d. External oblique fascia
   e. Transversalis fascia
5. Despite warnings from his friends and colleagues, a 32-year-old urologist purchases a motorcycle. He is involved in an accident and suffers grade 2 renal trauma. The hematoma would most likely travel in which direction if it continued to expand?
   a. Superior
   b. Lateral
   c. Medial
   d. Caudal
   e. Cephalad
6. The anterior and posterior laminae of Gerota fascia merge laterally to form:
   a. transversus abdominis.
   b. lumbodorsal fascia.
   c. lateral renal fascia.
   d. lateroconal fascia.
   e. perirenal fascia.
7. The kidneys are associated with which embryologic structure?
   a. Outer stratum
   b. Intermediate stratum
   c. Inner stratum
   d. Ectoderm
   e. Endoderm
8. Which of the following statements are TRUE?

a. Small mesenteric defects have a lower risk of internal hernia compared to large defects.
b. Based on current understanding, the mesentery is continuous.
c. In addition to its gastrointestinal (GI) function, the mesentery has distinct immunologic, endocrine, and vascular functions.
d. a and b.
e. a and c.
f. b and c.
g. a, b, and c.

9. The blood supply to the adrenal gland may include branches from the:
   a. inferior phrenic artery.
   b. aorta.
   c. renal artery.
   d. a and b.
   e. a and c.
   f. a, b, and c.
10. Which of the following statements is TRUE?
   a. The superior mesenteric artery (SMA) may be sacrificed without causing bowel ischemia.
   b. Ligation of the inferior mesenteric artery (IMA) will cause ischemia to the large bowel but not the small bowel.
   c. The IMA may be sacrificed without colonic ischemia because of collateral circulation via the marginal artery and hemorrhoidal arteries.
   d. The IMA may be sacrificed without colonic ischemia because of collateral circulation via the ileocolic artery.
   e. Neither the superior nor the inferior mesenteric arteries may be sacrificed without causing bowel ischemia.
11. Following colon mobilization for right-sided transperitoneal retroperitoneal surgery, which plane should be developed prior to approaching the renal hilum?
   a. Medial to the gonadal vein
   b. Posterior to the gonadal vein
   c. Lateral to the gonadal vein
   d. Medial to the inferior vena cava (IVC)

12. Which of the following statements is FALSE?
    a. The right testicular vein typically drains into the IVC.
    b. The left testicular vein typically drains into the left renal vein.
    c. Unilateral varicoceles are more common on the left side.
    d. A sudden-onset unilateral right-sided varicocele should prompt retroperitoneal imaging.
    e. The left ovarian vein typically drains into the IVC.
13. Which of the following statements is TRUE?
    a. The proximal ureter receives its blood supply medially, and the distal ureter receives its blood supply medially.
    b. The proximal ureter receives its blood supply medially, and the distal ureter receives its blood supply laterally.
    c. The proximal ureter receives its blood supply laterally, and the distal ureter receives its blood supply medially.
    d. The proximal ureter receives its blood supply laterally, and the distal ureter receives its blood supply laterally.
14. Which of the following is NOT a tributary of the splenic vein?
    a. Inferior mesenteric vein
    b. Short gastric vein
    c. Left gastroepiploic vein
    d. Right gastroepiploic vein
    e. Pancreatic veins
15. What statement best describes the lymphatic drainage of the right testis?
    a. Superficial then deep right inguinal nodes
    b. Left para-aortic with some drainage to the interaortocaval nodes
    c. Only to the interaortocaval nodes
    d. Primarily to the interaortocaval nodes with some drainage to the right paracaval nodes
    e. Interaortocaval nodes primarily with some drainage to the right paracaval nodes and a small but appreciable amount of drainage to the left para-aortic nodes
16. In addition to cranial nerves III, VII, IX, and X, parasympathetic nervous system outflow includes:
    a. preganglionic fibers from L3 to L4.

b. preganglionic fibers from T1 to L2.
c. preganglionic fibers from S2 to S4.
d. a and c.
e. b and c.
f. a, b, and c.

17. Which of the following statements is FALSE about the genitofemoral nerve?
    a. One branch provides sensation to the upper anterior thigh.
    b. The motor component of one branch allows for contraction of muscle during the cremasteric reflex.
    c. It may be injured during a psoas hitch procedure.
    d. It originates from the L3 to L4 segments.
    e. Sensation to the anterior scrotum in males and the mons pubic and labium majus in females.
18. Damage to the hypogastric plexuses during retroperitoneal dissection may result in:
    a. stress incontinence.
    b. erectile dysfunction.
    c. retrograde ejaculation.
    d. varicocele.
    e. autonomic dysreflexia.
19. What is the major function of the muscles innervated by the obturator nerve?
    a. Hip adduction
    b. Hip abduction
    c. Hip flexion
    d. Hip extension
    e. Knee flexion

Answers

1. **e. Transverse colon.** The contents of the retroperitoneum include the kidneys, ureters, adrenals, pancreas, second and third portions of the duodenum, ascending colon, descending colon, arterial structures including the aorta and its branches, venous structures

including the IVC and its tributaries, lymphatics, lymph nodes, sympathetic trunk, and lumbosacral plexus. The transverse colon is considered to be an intraperitoneal structure.

2. **a. Iliacus.** Psoas major functions in flexion of the thigh at the hip joint and is innervated by the anterior rami of L1, L2, and L3. Iliacus is the only muscle listed that also functions in flexion of the thigh at the hip joint.
3. **a. The intercostal neurovascular bundle travels between the internal intercostal and innermost intercostal muscles.** Fracture of the lower ribs should lead to a high clinical suspicion for injury to the retroperitoneal structures. The 11th and 12th ribs must be distinguished from the other ribs because they have no anterior connection with the sternum and are often referred to as floating ribs. The 10th rib is not a floating rib. These ribs are of clinical significance during palpation for the marking of a surgical incision. Bony landmarks are of critical importance for surgical incision planning, as they do not typically move. Landmarks like the umbilicus are less useful as they can change location depending on the positioning (e.g., an obese patient with a large pannus in flank position). The intercostal vessels and nerves travel between the internal intercostal and innermost intercostal muscles within the costal groove on the caudal margin of the superior rib.
4. **e. Transversalis fascia.** The transversalis fascia lies deep to the transversus abdominis muscle and superficial to the preperitoneal fat and peritoneum.
5. **d. Caudal.** The perirenal space around the kidney is cone-shaped and is open at its inferior extent in the extraperitoneal pelvis. If a hematoma were to form within the Gerota fascia, it would be able to travel in a caudal direction.
6. **d. Lateroconal fascia.** The anterior and posterior laminae of Gerota fascia merge laterally to form the lateroconal fascia, which functions to separate the anterior and posterior pararenal spaces. It can be visualized radiographically on computed tomographic (CT) scan and continues anterolaterally deep to the transversalis fascia.
7. **b. Intermediate stratum.** The kidneys arise from mesoderm. The outer stratum covers the epimysium of the abdominal wall muscles

and becomes the transversalis fascia. The intermediate stratum is associated with the urinary organs, and the inner stratum is associated with the gastrointestinal organs.

8. **f. b and c.** Knowledge of mesenteric anatomy and function is important for urologist. The mesentery is known to be continuous based on current understanding and it has GI, immunologic, endocrine, and vascular functions. Smaller mesenteric defects have a higher risk of internal hernia than large defects. As such, smaller defects should be closed primarily.
9. **f. a, b, and c.** The adrenal gland may receive branches from the superior adrenal artery off of the inferior phrenic artery, the middle adrenal artery off of the aorta, and the inferior adrenal artery off of the renal artery.
10. **c. The IMA may be sacrificed without colonic ischemia because of collateral circulation via the marginal artery and hemorrhoidal arteries.** The SMA supplies the pancreas (inferior pancreaticoduodenal artery), small intestine, and the majority of the large intestine (ileocolic, right colic, and middle colic). Ligation of the SMA will result in catastrophic bowel ischemia (without pancreatic ischemia because of collaterals from the celiac artery and the superior pancreaticoduodenal artery). The branches of the IMA are the left colic, sigmoid, and superior hemorrhoidal (rectal) arteries. The collateral circulation of the sigmoid artery via the marginal artery of Drummond and the inferior and middle hemorrhoidal arteries allows for the IMA to be sacrificed without colonic ischemia.
11. **c. Lateral to the gonadal vein.** The next major step following colon mobilization is identification of the ureter. During this step on the right, the plane is developed lateral to the gonadal which leaves the gonadal vein in a medial position. If the plane medial to the gonadal vein is developed, there is a risk of injury to the IVC, and with elevation of the kidney, avulsion of the gonadal may occur.
12. **e. The left ovarian vein typically drains into the IVC. The venous drainage of the ovarian and testicular veins is similar.** The right testicular and the right ovarian veins typically drain into

the IVC while the left testicular and the left ovarian veins drain into the left renal vein. Unilateral varicoceles are more common on the left, which may be a result of the increased length and perpendicular entry of the left testicular vein into the left renal vein. Given the rarity of unilateral right-side varicocele, a sudden-onset right-side varicocele should increase suspicion for a renal or retroperitoneal malignancy, leading to poor outflow and warrants retroperitoneal imaging.

13. **b. The proximal ureter receives its blood supply medially and the distal ureter receives its blood supply laterally.** In general, the blood supply of the ureter is medial proximally and lateral for the distal ureter. The renal artery and aorta supply the proximal ureter while the internal iliac artery, superior vesical, and inferior vesical arteries supply the distal ureter.
14. **d. Right gastroepiploic vein.** The tributaries of the splenic vein are the inferior mesenteric, short gastric, left gastroepiploic, and pancreatic veins. The right gastroepiploic vein is a tributary of the superior mesenteric vein.
15. **e. Interaortocaval nodes primarily with some drainage to the right paracaval nodes and a small but appreciable amount of drainage to the left para-aortic nodes.** The right testis drains primarily to the interaortocaval nodes, with some drainage to the right paracaval nodes. The left para-aortic region does receive a small but appreciable amount of lymphatic drainage from the right testis. This drainage pattern is consistent with the global lymphatic flow from right to left.
16. **c. Preganglionic fibers from S2 to S4.** The parasympathetic nervous system has craniosacral outflow with preganglionic fibers originating from cranial nerves III, VII, IX, and X, as well as the ventral rami of the second, third, and fourth sacral nerves.
17. **d. It originates from the L3 to L4 segments.** Each statement is true except for D. The genitofemoral nerve originates at the L1 to L2 segments.
18. **c. Retrograde ejaculation.** Much of the sympathetic innervation to the pelvic viscera travels through the superior and inferior hypogastric plexuses, which are contiguous. The superior

hypogastric plexus originates at the caudal extent of the abdominal aorta and extends to the anterior surface of the fifth lumbar vertebra. Extensive retroperitoneal dissection that causes disruption of these plexuses may result in loss of seminal vesicle emission or failure of bladder neck closure, resulting in retrograde ejaculation.

19. **a. Hip adduction.** The obturator nerve innervates the muscles of the medial thigh compartment. These include the gracilis, adductor longus, adductor brevis, adductor magnus, and obturator externus muscles. The muscles function to adduct and rotate the thigh at the hip joint.

TABLE 75.1

ARTERY	BRANCH	ORIGIN	SUPPLIES
Celiac trunk	Anterior	Immediately inferior to aortic hiatus of diaphragm	Abdominal foregut
Superior mesenteric artery	Anterior	Immediately inferior to celiac trunk	Abdominal midgut
Inferior mesenteric artery	Anterior	Inferior to renal arteries	Abdominal hindgut
Middle adrenal arteries	Lateral	Immediately superior to renal arteries	Adrenal glands
Renal arteries	Lateral	Immediately inferior to superior mesenteric artery	Kidneys
Testicular or ovarian arteries	Paired anterior	Inferior to renal arteries	Testes in male and ovaries in female
Inferior phrenic arteries	Paired lateral	Immediately inferior to aortic hiatus	Diaphragm
Lumbar arteries	Posterior	Usually four pairs	Posterior abdominal wall and spinal cord
Median sacral arteries	Posterior	Just superior to aortic bifurcation, pass inferiorly across lumbar vertebrae, sacrum, and coccyx	
Common iliac arteries	Terminal	Bifurcation usually occurs at the level of L4 vertebra	

Modified from Drake RL, Vogl W, Mitchell AWM: *Gray's anatomy for students,* Philadelphia, 2005, Elsevier, p 331.

TABLE 75.2

BRANCH	ORIGIN	SPINAL SEGMENTS	MOTOR FUNCTION	SENSORY FUNCTION
Subcostal	Anterior ramus T12	T12	Muscles of the abdominal wall	Skin over the hip
Iliohypogastric	Anterior ramus L1	L1	Internal oblique and transversus abdominis	Posterolateral gluteal skin and skin in pubic region
Ilioinguinal	Anterior ramus L1	L1	Internal oblique and transversus abdominis	Skin in the upper medial thigh, and either the skin over the root of the penis and anterior scrotum or the mons pubis and labium majus
Genitofemoral	Anterior rami L1 and L2	L1, L2	*Genital branch*: male cremasteric muscle	*Genital branch*: skin of anterior scrotum or skin of mons pubis and labium majus *Femoral branch*: skin of upper anterior thigh
Lateral cutaneous nerve of the thigh	Anterior rami L2 and L3	L2, L3	None	Skin on anterior and lateral thigh to the knee
Obturator	Anterior rami L2–L4	L2–L4	Obturator externus, pectineus, and muscles in medial compartment of thigh	Skin on medial aspect of the thigh
Femoral	Anterior rami L2–L4	L2–L4	Iliacus, pectineus, and muscles in anterior compartment of thigh	Skin on anterior thigh and medial surface of leg

Modified from Drake RL, Vogl W, Mitchell AWM: *Gray's anatomy for students,* Philadelphia, 2005, Elsevier, p 340.

Chapter Review (see also TABLES 75.1 and 75.2)

Retroperitoneum and Posterior Abdominal Wall

1. Retroperitoneal contents include the kidneys, ureters, adrenals, pancreas, portions of the duodenum, ascending colon, descending colon, mesentery, vasculature, lymphatics, and nervous structures.
2. The retroperitoneum is contained anteriorly by the posterior reflection of the peritoneum and posteriorly by the abdominal wall.
3. It is contained cranially and caudally by the diaphragm and the extraperitoneal pelvic structures, respectively.
4. The intercostal vessels and nerves travel between the internal intercostal and innermost intercostal muscles within the costal groove on the caudal margin of the superior rib.
5. The lumbodorsal fascia merges anterolaterally with the transversus abdominis muscle and is composed of three layers that cover the

posterior abdominal wall musculature.

Retroperitoneal Fasciae and Spaces

1. The boundaries of the posterior pararenal space are the posterior lamina of Gerota fascia anteriorly and the transversalis fascia posteriorly and laterally.
2. The anterior and posterior laminae of Gerota fascia form the boundaries of the perirenal space which has a conelike shape that is open caudally in the extraperitoneal pelvis.
3. Perinephric fluids collections can expand caudally due to this opening in the perirenal space.
4. The white line of Toldt represents the lateral border of the fusion of the colonic mesentery with the posterior peritoneum.

Vasculature

1. The blood supply to the adrenal arises from the inferior phrenic, the aorta, and the renal artery.
2. The renal hila are at the level of L1, and the renal veins are anterior to the renal arteries.
3. The paired gonadal arteries typically arising anterolaterally from the aorta below the renal arteries
4. The SMA to IMA collateral circulation occurs via the marginal artery of Drummond, which allows the IMA to be sacrificed without colonic ischemia.
5. In general, the proximal ureter receives its blood supply medially, and the distal ureter receives its blood supply from a lateral direction.
6. The left gonadal vein enters left renal vein at a right angle and the right gonadal vein enters into the IVC directly; this anatomic distinction results in higher rates of varicoceles on the left.

Lymphatic System

1. Testes are embryologically retroperitoneal and have retroperitoneal blood supply and primary lymphatic drainage.
2. The lymph of the left testis drains to the left para-aortic nodes with some drainage to the interaortocaval nodes.
3. The right testis drains primarily to the interaortocaval nodes with some to the right paracaval nodes and a small amount to the left para-aortic region.
4. The lymphatic drainage of the testicles is consistent with global lymphatic flow from right to left.

Nervous System

1. The parasympathetic autonomic nervous system has craniosacral outflow and the postganglionic fibers are often contained within the walls of the innervated viscera.
2. The preganglionic sympathetic nervous system fibers exit from the spinal cord from T1 to L2 and may synapse within the sympathetic trunk or within the autonomic plexuses.
3. The somatic nervous system provides sensory and motor innervation to the pelvis and lower extremities through the lumbosacral plexus.

76

Neoplasms of the Testis

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Keywords

carcinoma; embryonal; drug therapy; endodermal sinus tumor; lymph node excision; neoplasm recurrence; neoplasm staging; radiotherapy; seminoma; teratoma; testicular neoplasms; testis; watchful waiting

Questions

1. The following adult male germ cell tumor (GCT) subtypes arise from germ cell neoplasia in situ (GCNIS) EXCEPT:
    a. embryonal tumor.
    b. choriocarcinoma.
    c. classic seminoma.
    d. spermatocytic tumor.
    e. teratoma.
2. Which of the following statements is TRUE regarding spermatocytic tumor?
    a. Cryptorchidism is a risk factor.
    b. It may occur as a mixed GCT with other histologic GCT subtypes.
    c. It may contain i(12p) mutations.
    d. Bilateral testicular involvement may occur in 2% to 3% of cases.

e. Metastatic spermatocytic tumor is rare.

3. Which of the following GCT subtypes is most likely to spread hematogenously?
    a. Choriocarcinoma
    b. Embryonal carcinoma
    c. Immature teratoma
    d. Teratoma with malignant transformation
    e. Seminoma
4. A 24-year-old man presents with a solid, painless, right intratesticular mass confirmed by scrotal ultrasonography. His left testis is normal. Serum tumor markers show a human chorionic gonadotropin (hCG) value of 96 mU/mL (upper limit: <5 mU/mL) and an α-fetoprotein (AFP) value of 58 ng/mL (upper limit: <11 ng/mL). The most likely histologic finding in the right testis is:
    a. pure teratoma.
    b. pure seminoma.
    c. pure embryonal carcinoma.
    d. pure yolk sac tumor.
    e. choriocarcinoma.
5. Which of the following is an acceptable indication for testis-sparing surgery?
    a. 1.3-cm solid intratesticular mass with a normal contralateral testis
    b. Suspected benign testicular lesion
    c. 2.4-cm solid mass in a solitary testis
    d. Hypogonadal male with 1.2-cm solid intratesticular mass in a solitary testis
    e. Small (<1 cm) hyperechoic lesion suggestive of a “burned out” primary tumor in a patient with disseminated nonseminomatous GCT (NSGCT) with serum-elevated AFP and hCG
6. A 37-year-old man presents with a 5-cm left testicular mass. Computed tomography (CT) reveals a 6-cm para-aortic mass but no evidence of distant metastases. Serum tumor markers show an AFP level of 1100 ng/mL (upper limit: <11 ng/mL) and an hCG level of 80 mU/mL (upper limit: <5 mU/mL). Left inguinal orchiectomy

reveals a mixed GCT with 60% embryonal carcinoma, 30% yolk sac tumor, 5% seminoma, and 5% teratoma. The next best management step is:

a. retroperitoneal lymph node dissection (RPLND).
b. induction chemotherapy with three cycles of bleomycin-etoposide-cisplatin.
c. induction chemotherapy with four cycles of bleomycin-etoposide-cisplatin.
d. to obtain repeat serum tumor marker levels in 7 days.
e. CT-guided biopsy of the para-aortic mass.

7. All of the following patients would be classified as "poor-risk" by International Germ Cell Cancer Collaborative Group (IGCCCG) classification criteria EXCEPT those with:

a. testicular seminoma with brain metastases.
b. primary mediastinal NSGCT.
c. testicular NSGCT with rising postorchiectomy AFP of 15,000 ng/mL (upper limit: <11 ng/mL).
d. primary retroperitoneal NSGCT with liver metastases.
e. testicular NSGCT with rising postorchiectomy hCG of 93,000 mU/mL (upper limit: <5 mU/mL).

8. A 34-year-old African American man with a left testicular mass undergoes inguinal orchiectomy that reveals a 1.2-cm pure seminoma that is confined to the testis with no evidence of lymphovascular invasion or rete testis invasion. His postorchiectomy serum tumor markers are within the normal range. CT of the chest-abdomen-pelvis reveals no evidence of retroperitoneal lymphadenopathy and no evidence of pulmonary metastases. However, on the chest images, there is evidence of bulky hilar adenopathy bilaterally. The next best management step is:

a. induction chemotherapy with four cycles of bleomycin-etoposide-cisplatin.
b. induction chemotherapy with four cycles of etoposide-cisplatin.
c. mediastinoscopy and biopsy.
d. close observation.

e. bilateral thoracotomy and resection.

9. A 43-year-old man with clinical stage IIA left seminoma receives dog-leg radiation therapy to the retroperitoneum and ipsilateral pelvis with a boost to his solitary 2-cm para-aortic mass. Six months after completing treatment, surveillance CT reveals a persistent para-aortic mass that has now grown to 2.8 cm. The remainder of his metastatic evaluation is negative, and his serum tumor marker levels are all within normal limits. The next best management step is:
    a. RPLND.
    b. CT-guided biopsy of the retroperitoneal mass.
    c. close observation until the mass regresses or the patient develops distant metastases.
    d. induction chemotherapy with three cycles of bleomycin-etoposide-cisplatin.
    e. salvage chemotherapy with four cycles of paclitaxel-ifosfamide-cisplatin.
10. A 41-year-old man has GCNIS discovered on biopsy of an atrophic right testis during investigations for infertility due to azoospermia. He has a history of left inguinal hernia repair. His left testis is normal in size and consistency, and there is evidence of normal spermatogenesis on testicular biopsy. His serum luteinizing hormone (LH), follicle-stimulating hormone (FSH), and testosterone levels are within the normal range. The most appropriate treatment for the GCNIS in the right testis now is:
    a. inguinal orchiectomy.
    b. low-dose radiation therapy.
    c. carboplatin.
    d. observation.
    e. transscrotal orchiectomy.
11. Which of the following factors is NOT associated with the presence of occult metastases in clinical stage I NSGCT?
    a. Lymphovascular invasion
    b. Absence of yolk sac tumor in the primary tumor
    c. Percentage of embryonal carcinoma in the primary tumor
    d. Elevated preorchiectomy AFP level

e. Advanced primary tumor stage

12. A 27-year-old convict at a correctional facility presents for management of clinical stage I left NSGCT. He has a history of enlarging left testicular mass for 12 months that was discovered incidentally during a routine physical examination by the prison physician. Pathologic examination of the orchiectomy specimen revealed a 1.2-cm mixed GCT (40% seminoma, 40% embryonal carcinoma, 20% yolk sac tumor) confined to the testis without evidence of lymphovascular invasion. His postorchiectomy serum tumor markers are within normal limits. He has a history of multiple incarcerations in the past, and his viral serology is positive for hepatitis C. The most appropriate treatment is:
    a. adjuvant radiation therapy to the retroperitoneum and ipsilateral pelvis.
    b. surveillance.
    c. chemotherapy with one cycle of bleomycin-etoposide-cisplatin.
    d. chemotherapy with two cycles of carboplatin.
    e. RPLND.
13. Which of the following factors is NOT associated with the presence of necrosis/fibrosis in residual masses after first-line chemotherapy?
    a. Absence of teratoma in the primary tumor.
    b. Residual mass size.
    c. Percentage shrinkage of mass after chemotherapy.
    d. Prechemotherapy mass size.
    e. Lymphovascular invasion.
14. A 37-year-old man presents for treatment of a 1.2-cm left testicular mixed GCT (40% teratoma, 40% seminoma, 15% embryonal carcinoma, 5% yolk sac tumor) confined to the testis without evidence of lymphovascular invasion. His postorchiectomy serum tumor marker levels are within normal limits. Chest CT shows no evidence of metastatic disease. Abdominopelvic CT shows a 7-mm nodule in the paracaval location just inferior to the right renal hilum. The remainder of the CT study is unremarkable. His

medical history is also unremarkable. The most appropriate management is:

a. CT-guided biopsy of the paracaval lesion.
b. RPLND.
c. two cycles of chemotherapy with bleomycin-etoposide-cisplatin.
d. observation.
e. three cycles of chemotherapy with bleomycin-etoposide-cisplatin.

15. The following factors are associated with the presence of occult distant metastases in patients with clinical stage IIA-B NSGCT EXCEPT:
    a. elevated postorchiectomy hCG.
    b. lymphovascular invasion.
    c. retroperitoneal mass size.
    d. large primary tumor with involvement of the scrotal skin.
    e. retroperitoneal lymphadenopathy outside the primary landing zone.
16. The following are independent risk factors for relapse postchemotherapy RPLND EXCEPT:
    a. evidence of viable malignancy in resected specimens.
    b. incomplete resection.
    c. rising pre-RPLND serum tumor markers.
    d. poor-risk disease at diagnosis by IGCCCG criteria.
    e. prior RPLND.
17. A 34-year-old man with right clinical stage III NSGCT (100% embryonal carcinoma) with good-risk features by IGCCCG criteria receives induction chemotherapy with three cycles of bleomycin-etoposide-cisplatin. At completion of chemotherapy, his serum tumor markers are within normal limits. On postchemotherapy CT studies he has a 1.7-cm mass (4.8 cm at diagnosis) in the interaortocaval region and a 0.8-cm mass in the para-aortic region (2.3 cm at diagnosis). He also has bilateral pulmonary nodules in the right lower lobe (0.6 cm; 1.4 cm at diagnosis) and left upper lobe (0.8 cm; 1.6 cm at diagnosis). The most appropriate management is:

a. four cycles of vinblastine-ifosfamide-cisplatin second-line chemotherapy.
b. resection of the interaortocaval mass.
c. bilateral postchemotherapy RPLND.
d. bilateral thoracotomy and resection of residual pulmonary masses.
e. CT-guided biopsy of the pulmonary mass(es).

18. Which of the following statements is FALSE concerning late relapse of NSGCT?
    a. Surgical resection is the primary treatment modality.
    b. Yolk sac tumor is the most common malignant histology.
    c. The incidence is increasing.
    d. The retroperitoneum is the most common site.
    e. The outcome is poor relative to those with early NSGCT relapse.
19. A 35-year-old man with clinical stage IIC left mixed GCT (50% embryonal, 40% teratoma, 10% yolk sac) with good-risk features by IGCCCG criteria receives three cycles of bleomycin-etoposide-cisplatin chemotherapy. At the start of chemotherapy, his AFP was 380 ng/mL (upper limit: <11 ng/mL), and this has normalized at the end of chemotherapy. Restaging CT shows the solid para-aortic mass has increased from 5.3 cm to 8.9 cm with displacement of the aorta and left kidney as well as new lymphadenopathy in the left common iliac and left obturator region. The patient complains of recent onset of left-sided back pain. The most appropriate management is:
    a. RPLND and pelvic lymph node dissection.
    b. CT-guided biopsy of the para-aortic mass.
    c. four cycles of paclitaxel-ifosfamide-cisplatin as second-line chemotherapy.
    d. two cycles of bleomycin-etoposide-cisplatin followed by carboplatin-etoposide high-dose chemotherapy and autologous stem cell rescue.
    e. bleomycin-etoposide-cisplatin plus radiation therapy.
20. The rationale for single-agent carboplatin as treatment for clinical stage I seminoma is based on all of the following factors EXCEPT:

a. absence of teratoma.
b. less neurotoxicity compared with cisplatin.
c. less nephrotoxicity compared with cisplatin.
d. less ototoxicity compared with cisplatin.
e. similar efficacy to cisplatin.

21. Late complications of infradiaphragmatic dog-leg radiotherapy include all the following EXCEPT:
    a. peptic ulcer disease.
    b. coronary artery disease.
    c. secondary malignancy.
    d. ejaculatory dysfunction.
    e. impaired spermatogenesis.
22. The rationale for surveillance in clinical stage I seminoma is based on all of the following factors EXCEPT:
    a. Utility of serum tumor markers to identify relapse at an early and curable stage.
    b. Relapses are cured in virtually all cases by deferred dog-leg radiotherapy.
    c. Lack of validated histopathologic prognostic factors to identify a high-risk subset.
    d. Improved short- and long-term toxicity compared with primary radiotherapy and carboplatin.
    e. 15% to 20% of patients are cured by orchiectomy.
23. A 44-year-old man with clinical stage III left testicular seminoma with IGCCCG good-risk features has a discrete 2.4-cm residual para-aortic mass (3.8 cm at diagnosis) after receiving three cycles of bleomycin-etoposide-cisplatin chemotherapy. His pulmonary nodules have regressed completely. His serum tumor markers are within the normal range. The most appropriate management is:
    a. postchemotherapy radiation therapy to the residual mass.
    b. fluorodeoxyglucose-labeled positron emission tomography (FDG-PET) at least 4 weeks after completing chemotherapy.
    c. observation.
    d. postchemotherapy surgical resection of the residual mass.
    e. Four cycles of paclitaxel-ifosfamide-cisplatin as second-line chemotherapy.

24. A 42-year-old asymptomatic man presents for management of right NSGCT (80% embryonal carcinoma, 10% teratoma, 10% choriocarcinoma). His preorchiectomy hCG value was 15,000 mU/mL (upper limit: <5 mU/mL), and this has risen to 50,800 mU/mL after orchiectomy. Chest CT shows numerous pulmonary nodules. There is evidence of multiple masses in the interaortocaval region (largest, 4.8 cm) and masses in the para-aortic region (largest, 2.6 cm). The most appropriate management is:
    a. three cycles of bleomycin-etoposide-cisplatin chemotherapy.
    b. RPLND.
    c. four cycles of bleomycin-etoposide-cisplatin chemotherapy.
    d. CT of the head.
    e. two cycles of bleomycin-etoposide-cisplatin followed by carboplatin-etoposide high-dose chemotherapy and autologous stem cell rescue.
25. Which of the following statements is FALSE regarding treatment-related toxicity?
    a. Two cycles of platin-based chemotherapy do not increase one's risk of developing cardiovascular disease or secondary malignant neoplasm (SMN).
    b. Frequent CT body imaging may increase the risk of SMN.
    c. The risk of cardiovascular disease is highest among patients receiving mediastinal radiotherapy.
    d. Exposure to cisplatin-based chemotherapy and history of cigarette smoking are associated with similar risks of cardiovascular disease and SMN.
    e. Suprahilar dissection, vascular reconstruction, and hepatic resection are risk factors for chylous ascites after RPLND.
26. Which of the following are NOT similarities between Leydig cell tumors and GCT?
    i. Both are associated with a history of cryptorchidism.
    ii. Radical inguinal orchiectomy is the initial treatment of choice.
    iii. Bilateral tumors occur in 2% to 3% of cases.
    iv. Both may be associated with gynecomastia.

v. The retroperitoneum is the most common site of metastatic disease.

a. i, ii, and iii.
b. i and iii.
c. i, ii, iii, and iv.
d. v only.
e. All the above.

27. A 54-year-old man presents with an enlarging right inguinal mass. On examination, a palpable mass is noted in the right inguinal region that extends into the right hemiscrotum. The testis cannot be distinguished from this mass. Staging CT reveals a heterogeneous, infiltrative area of low-intensity mass (−20 Hounsfield units), 6 × 9 cm, involving the right spermatic cord and extending from the inguinal canal into the scrotum with displacement of the right testis. There is no evidence of retroperitoneal lymphadenopathy or distant metastases. The most appropriate management is:
    a. inguinal orchiectomy followed by adjuvant radiotherapy.
    b. inguinal orchiectomy alone.
    c. transscrotal orchiectomy.
    d. inguinal orchiectomy followed by ifosfamide-based adjuvant chemotherapy.
    e. inguinal orchiectomy followed by RPLND.

Pathology

1. A 26-year-old man has a right radical orchiectomy for an embryonal carcinoma of the testis. At the time of surgery, a contralateral biopsy is performed and reveals intratubular germ cell neoplasia (Fig. 76.1). The patient should be advised that he:
    a. should have a radical orchiectomy.
    b. has a significant chance of developing a germ cell tumor in the left testis.
    c. should not try to have a child.
    d. should immediately receive radiation to the testis.
    e. should receive salvage chemotherapy.

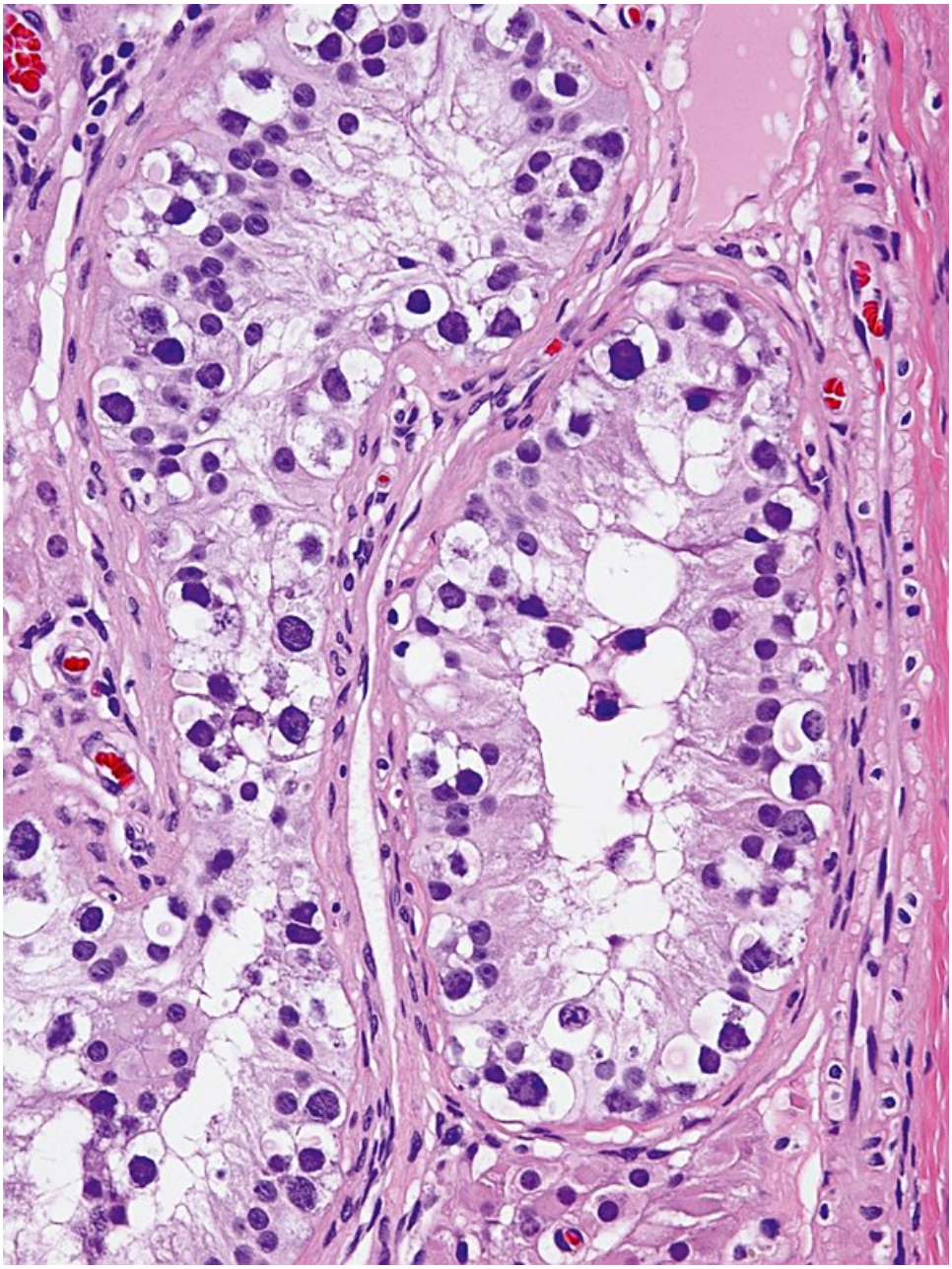


FIG. 76.1
From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Edinburgh, 2008, Mosby.

2. A 35-year-old man has an asymptomatic right scrotal mass. Testicular ultrasonography reveals a 3-cm heterogeneous intratesticular mass. A right radical orchiectomy is performed. The histology is depicted in Fig. 76.2 and is reported as seminoma. Abdominal CT scan is normal. The patient should be advised to:

a. receive radiation to the contralateral testis.
b. receive at least four cycles of chemotherapy.
c. be advised that observation is not an option.
d. be advised to have radiation therapy to the retroperitoneum.
e. receive radiation to the abdomen and chest.

3. A 32-year-old man has a right radical orchiectomy for a testicular mass. Preoperatively, his AFP value was normal and his hCG level was elevated at 5000 units. The histology is depicted in Fig. 76.3 and is reported as seminoma with giant cells. The next step in management is:
   a. follow markers and check half-life.
   b. chemotherapy according to choriocarcinoma protocol.
   c. RPLND.
   d. radiation therapy to retroperitoneum.
   e. three cycles of chemotherapy.
4. A 50-year-old man has a right radical orchiectomy for a testicular mass. The histology is depicted in Fig. 76.4 and is a spermatocytic tumor. Abdominal and chest CT are negative. Serum markers are normal. The patient should be advised to:
   a. receive radiation to the retroperitoneum.
   b. receive one cycle of chemotherapy.
   c. have a biopsy of the contralateral testis.
   d. not have any treatment.
   e. have a PET-CT scan.
5. A 20-year-old man has a right radical orchiectomy. The pathology is depicted in Fig. 76.5 and is read as embryonal carcinoma. His hCG and AFP values are elevated and a CT of abdomen and chest reveals no evidence of metastatic disease. Three weeks later, repeat AFP and hCG testing show no change in either marker. The patient should be advised to:
   a. have induction chemotherapy.

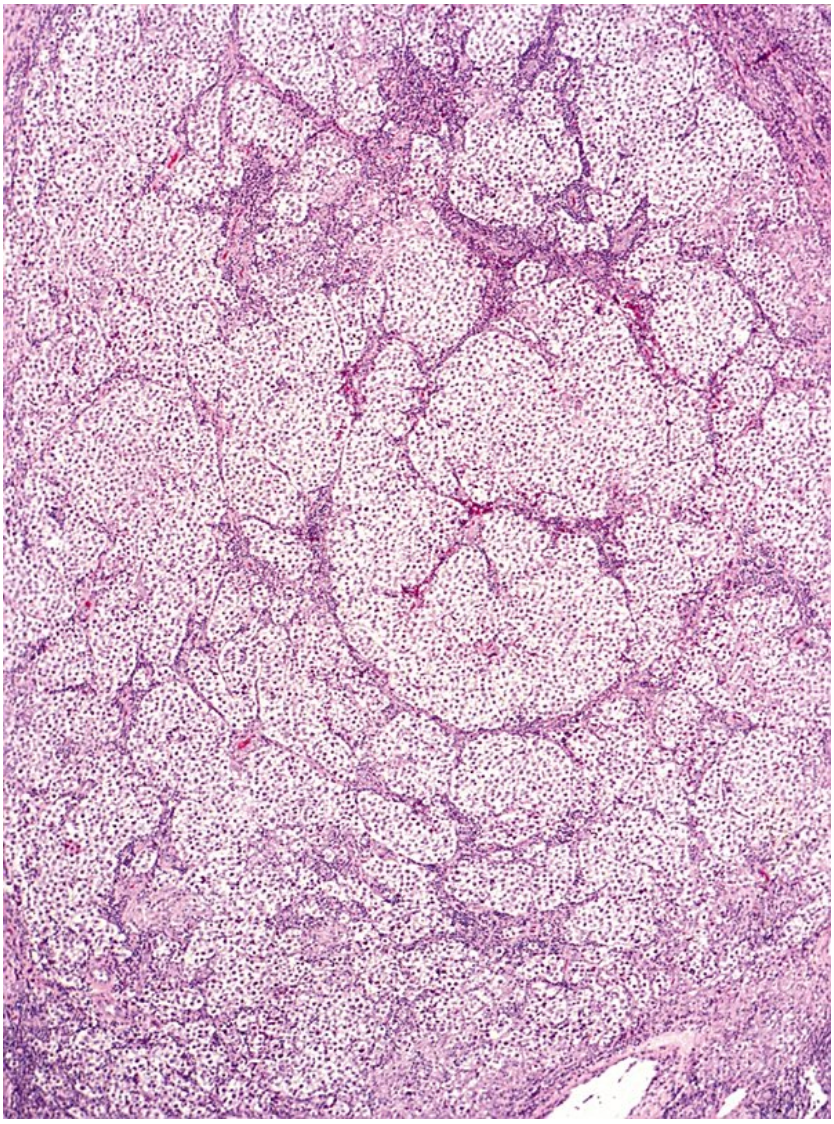


FIG. 76.2

From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Edinburgh, 2008, Mosby.

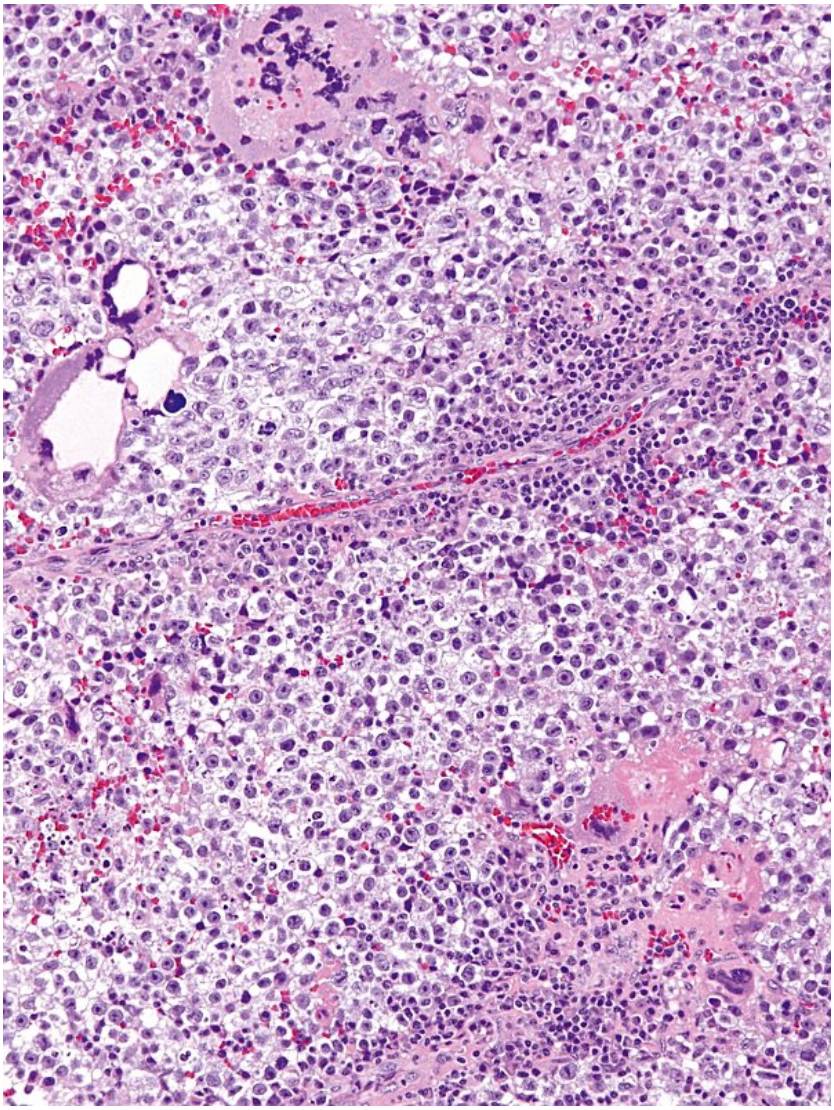


FIG. 76.3

From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Edinburgh, 2008, Mosby.

b. have an RPLND.
c. have a PET-CT.
d. receive radiotherapy below the diaphragm.
e. repeat the hCG and AFP tests in another month.

6. A 25-year-old man has a right radical orchiectomy. The histology is depicted in Fig. 76.6 and is reported as a mature teratoma. The patient's AFP is slightly elevated and bHCG is negative; however, there is a 3-cm mass in the retroperitoneum on CT. He is given

chemotherapy and the mass shrinks to 1.8 cm. The patient should be advised to:

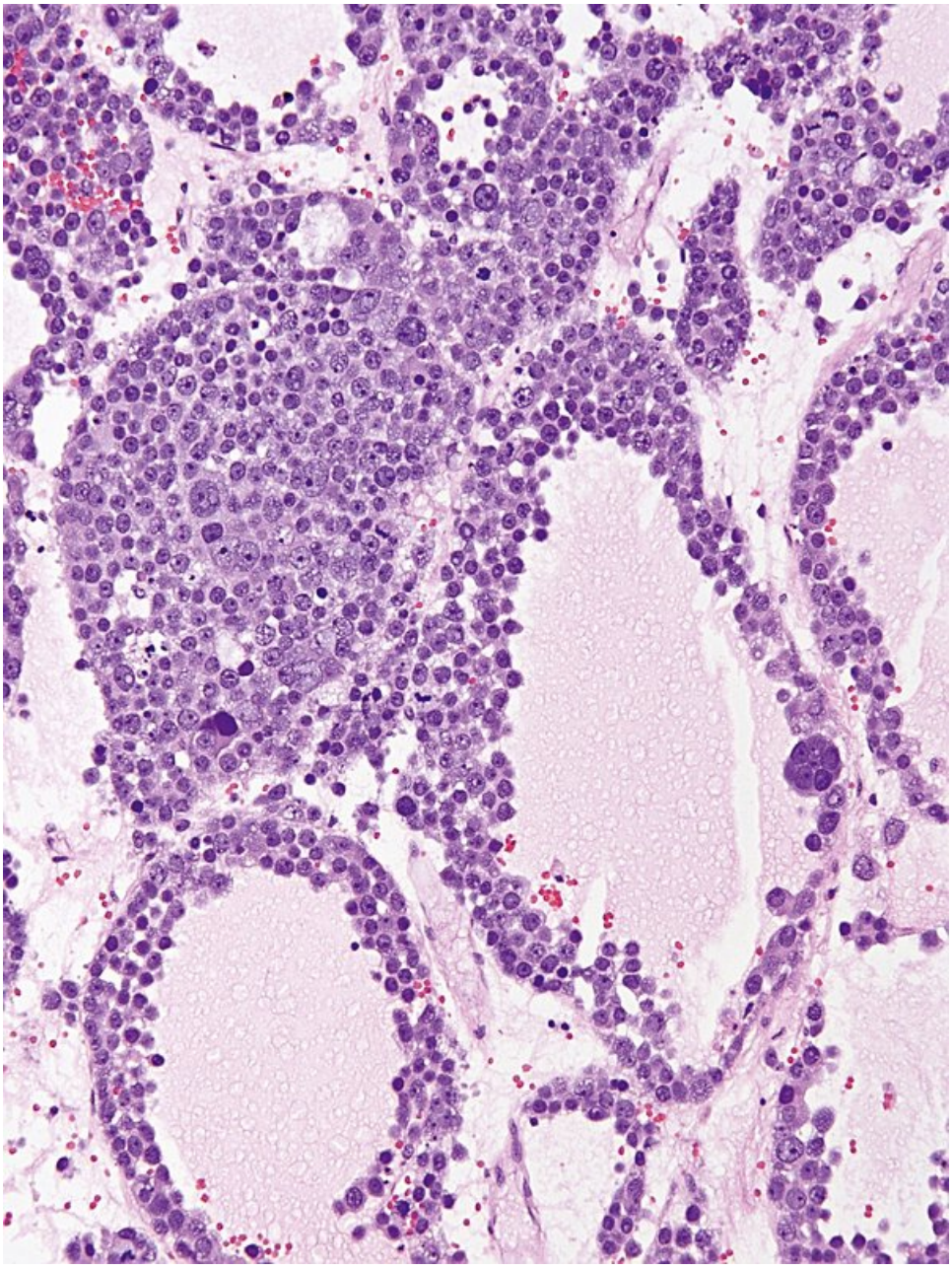


FIG. 76.4
From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Edinburgh, 2008, Mosby.

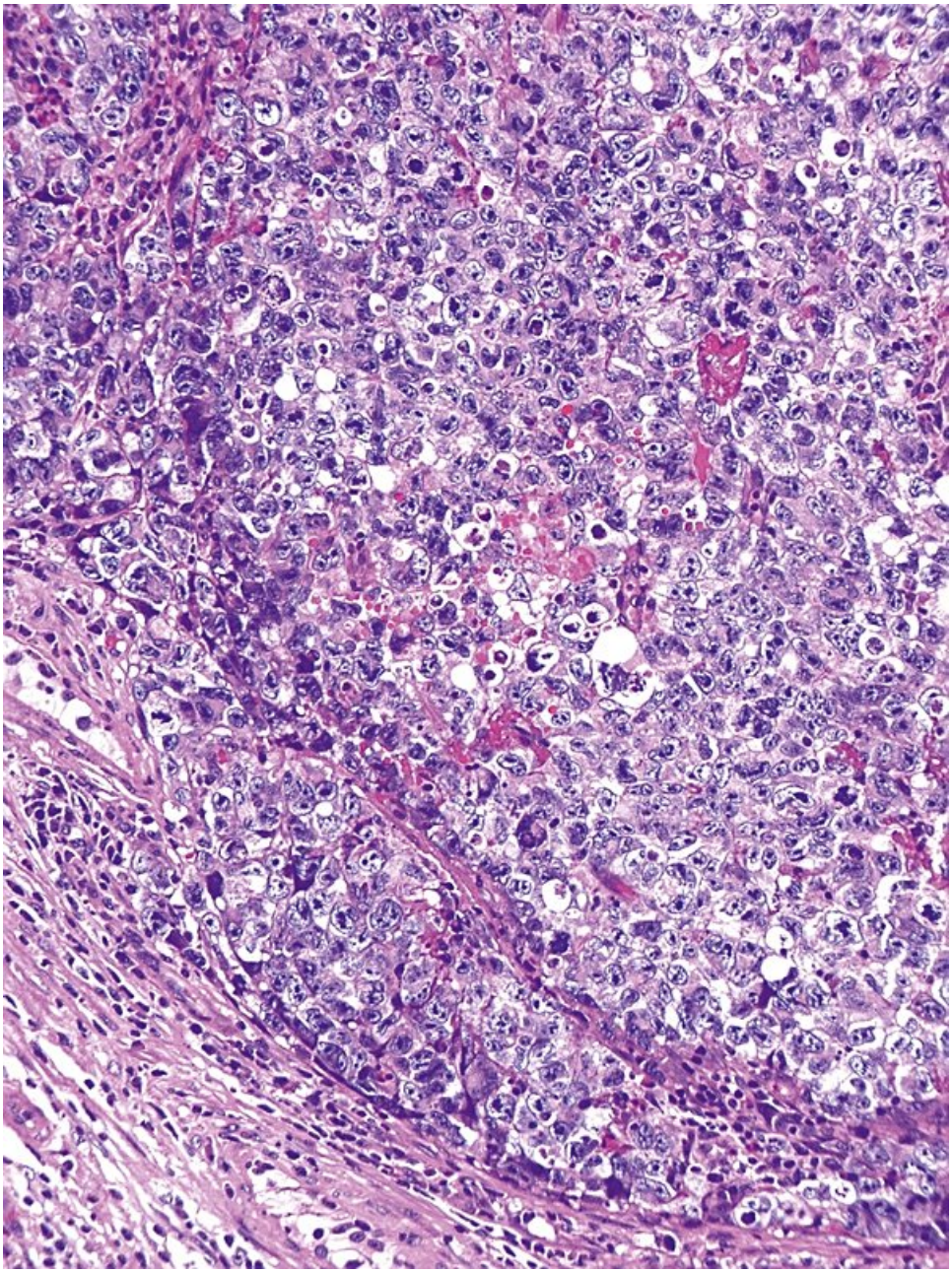


FIG. 76.5
From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Edinburgh, 2008, Mosby.

a. have a retroperitoneal lymphadenectomy dissection (RPLND).
b. have salvage chemotherapy.
c. get an FDG-PET scan.
d. receive radiation therapy.
e. be observed.

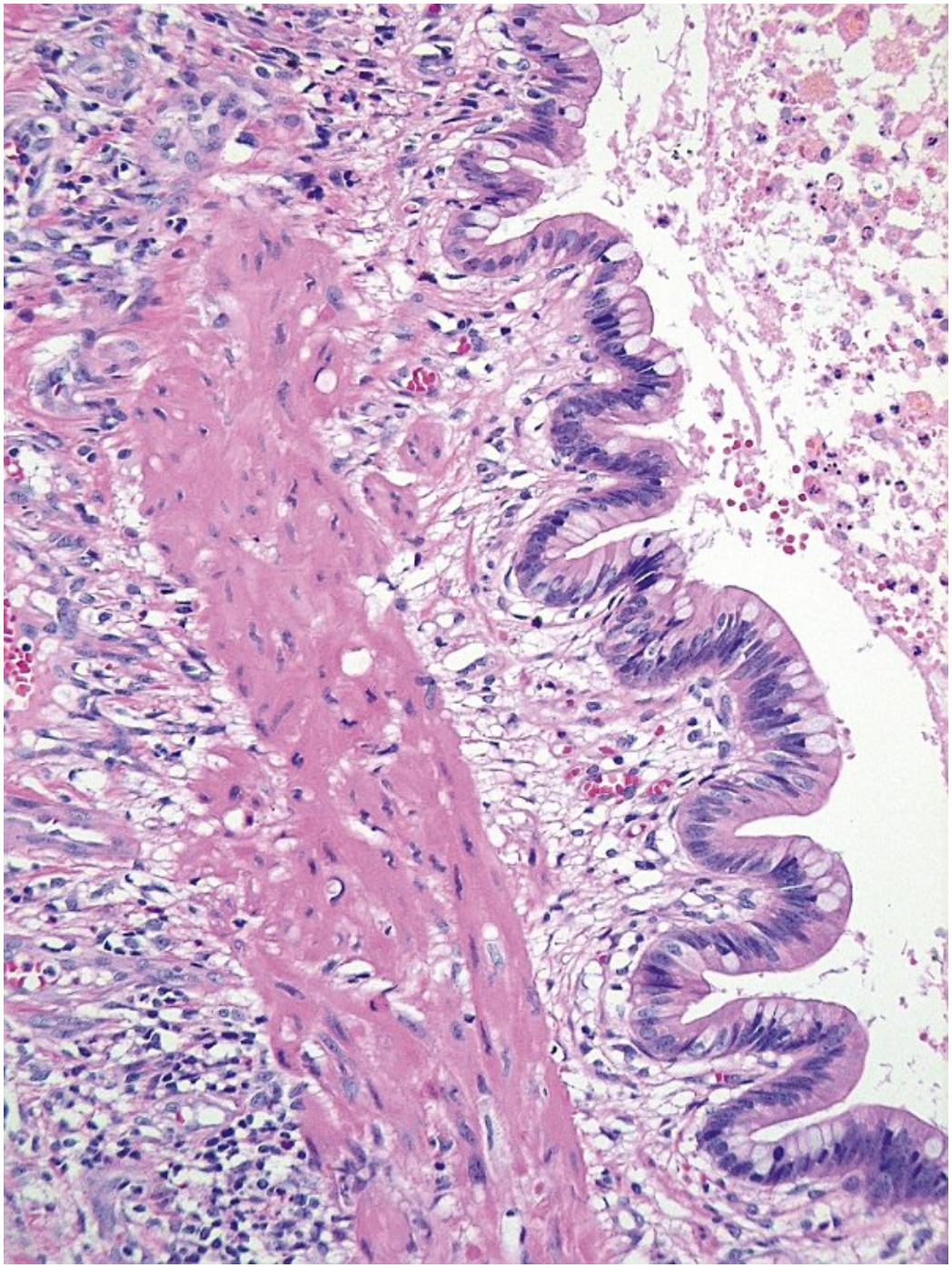


FIG. 76.6
From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Edinburgh, 2008, Mosby.

Imaging

1. A 36-year-old man noted a firm left scrotal mass. He was hit in the groin 1 month earlier with a tennis ball. Currently he has no pain, fever, or chills. The testicular ultrasound image is depicted in Fig. 76.7. The most likely diagnosis is:
    a. ruptured testis with peritesticular hematoma.
    b. testicular neoplasm.
    c. epidermoid cyst.
    d. dilated rete testis.
    e. testicular abscess.
2. A 32-year-old man had a left radical orchiectomy. Pathologic evaluation reveals a mixed GCT containing seminoma and embryonal cell carcinoma. Tumor markers are negative. The CT image depicted in Fig. 76.8 was obtained 1-day postoperation. Chest CT is negative. The next step in management is:
    a. biopsy.
    b. radiation therapy.
    c. chemotherapy.
    d. RPLND.
    e. repeat CT in 1 week to confirm a postsurgical inflammatory response.

Answers

1. **d. Spermatocytic tumor. GCNIS is the common precursor lesion for all types of adult male GCT with the exception of spermatocytic tumor.** Pediatric GCTs do not typically arise from GCNIS.
2. **e. Metastatic spermatocytic tumor is rare.** Spermatocytic tumor differs from other GCT subtypes in that it does not arise from GCNIS, cryptorchidism is not a risk factor, bilaterality has not been reported, it does not express i(12p) or placental alkaline phosphatase, and it does not occur as a mixed GCT with other GCT subtypes. Only one documented case of metastasis has been reported, and these lesions are almost always cured by orchiectomy.


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WF 38Hz
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+4.3
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cm/s
4.5
LEFT TESTICLE SAG L/M

FIG. 76.7

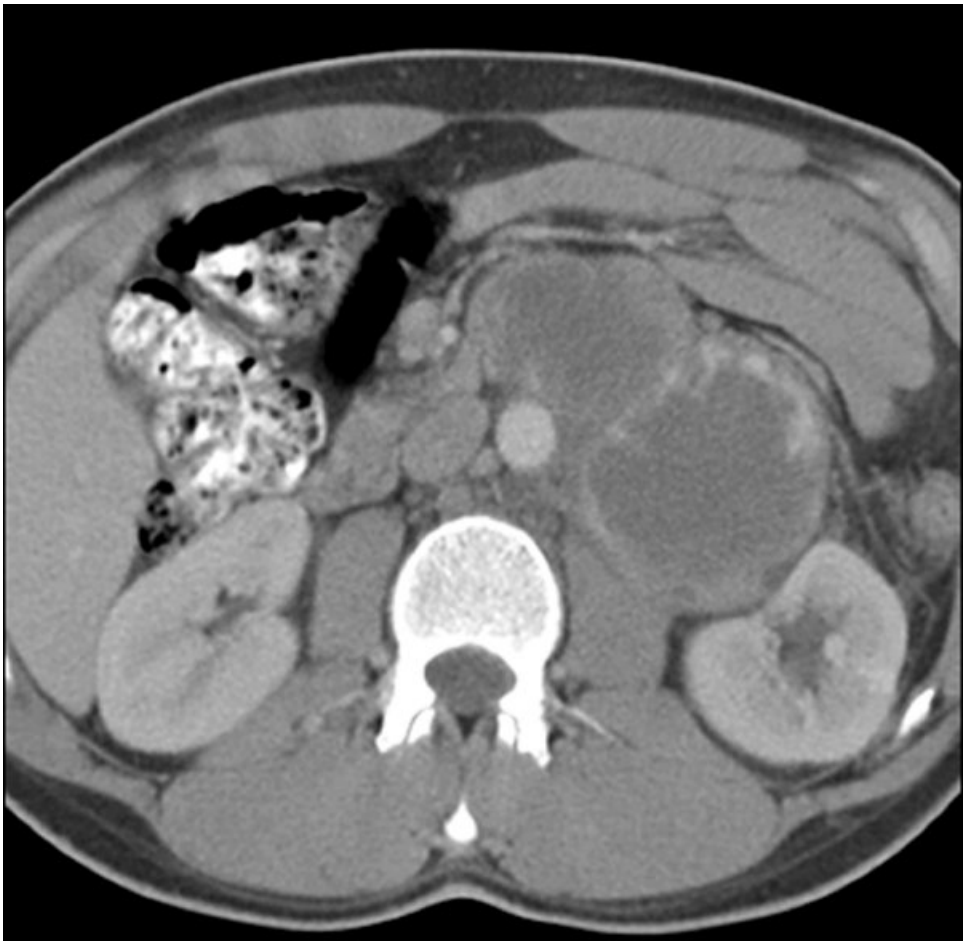


FIG. 76.8

3. **a. Choriocarcinoma.** With the exception of choriocarcinoma, the most common route of disease dissemination is via lymphatic channels from the primary tumor to the retroperitoneal lymph nodes and subsequently to distant sites. **Choriocarcinoma has a propensity for hematogenous dissemination. Yolk sac tumors in children are thought to spread hematogenously as well.**
4. **c. Pure embryonal carcinoma. Pure embryonal carcinoma may produce both AFP and hCG. Pure seminoma is associated with elevated serum hCG levels in 15% of cases but does not produce AFP. Pure teratoma typically is not associated with elevated serum tumor markers, although slightly elevated AFP levels may be observed. Choriocarcinoma is uniformly associated with elevated hCG levels but does not produce AFP.**

Most yolk sac tumors produce AFP, but they do not produce hCG.

5. **b. Suspected benign testicular lesion.** Testis-sparing surgery should be considered only in patients with suspected GCT who have normal testicular androgen production and who have a small (<2 cm) tumor either in a solitary testis or in the setting of bilateral synchronous testicular GCT. Testis-sparing surgery should not be performed in patients with suspected GCT who have a normal contralateral testis. Testis-sparing surgery may also be considered in patients with suspected benign testicular lesions such as an epidermoid cyst or adenomatoid tumor arising from the tunica albuginea.
6. **d. To obtain repeat serum tumor marker levels in 7 days.** Patients with elevated serum tumor markers before orchiectomy should have these levels measured after orchiectomy to assess whether the levels are declining, stable, or rising. Management decisions should not be made based on serum tumor marker levels before orchiectomy. Patients with rising postorchiectomy serum tumor marker levels should receive chemotherapy. The IGCCCG classification of metastatic NSGCT is based on the postorchiectomy serum tumor marker levels.
7. **a. Testicular seminoma with brain metastases.** According to IGCCCG classification criteria, there is no "poor risk" category for metastatic seminoma. Patients with metastatic seminoma who have nonpulmonary visceral metastases (e.g., liver, bone, brain) are classified as at intermediate risk. Metastatic NSGCT patients with mediastinal primary tumor *or* nonpulmonary visceral metastases *or* postorchiectomy AFP >10,000 ng/mL, *or* postorchiectomy hCG of 50,000 mU/mL are classified as at poor risk.
8. **c. Mediastinoscopy and biopsy.** The presence of distant metastasis in the absence of retroperitoneal disease or elevated serum levels of tumor markers is uncommon, particularly for patients with testicular seminoma. Therefore, these patients should undergo biopsy and histologic confirmation of the suspected lesion before management decisions are made.

9. **b. CT-guided biopsy of the retroperitoneal mass.** The risk of teratoma at metastatic sites is less of a consideration for metastatic seminoma than for NSGCT. Although rare, seminoma may transform into NSGCT elements, and this should be considered in patients with metastatic seminoma who fail to respond to conventional therapy. These patients should undergo biopsy and histologic confirmation of the suspected lesion before management decisions are made. Either an open or a laparoscopic biopsy of the para-aortic mass is an acceptable approach if CT-guided biopsy is not feasible or the result is nondiagnostic. However, an RPLND should not be performed without histologic confirmation of NSGCT pathology.
10. **a. Inguinal orchiectomy.** Inguinal orchiectomy and low-dose radiation therapy are associated with the highest rates of local control of GCNIS. In a patient with a normal contralateral testis (particularly if future paternity is desired), inguinal orchiectomy is the preferred choice owing to the deleterious effects of radiation therapy on spermatogenesis within the contralateral testis.
11. **d. Elevated preorchiectomy AFP level.** Preorchiectomy serum tumor marker levels are not associated with the presence of occult metastases in clinical stage I NSGCT. The presence of postorchiectomy serum tumor marker levels in clinical stage I NSGCT indicates the presence of occult systemic disease.
12. **c. Chemotherapy with one cycle of bleomycin-etoposide-cisplatin.** Chemotherapy is associated with the lowest risk of recurrence and is thus preferred versus surveillance for patients who are anticipated to be noncompliant with surveillance imaging and testing (even for patients at low risk for occult metastases). Chemotherapy and RPLND are associated with similar rates of long-term cure, but the former may be preferable in patients with transmissible diseases. Adjuvant radiation therapy and carboplatin are standard treatment approaches for clinical stage I seminoma.
13. **e. Lymphovascular invasion. Absence of teratoma in the primary tumor, prechemotherapy and postchemotherapy mass size, and percentage shrinkage of mass with chemotherapy are all associated with the presence of necrosis/fibrosis in residual**

masses after first-line chemotherapy. However, none of these factors (alone or together) is sufficiently accurate to exclude the presence of residual teratoma or viable malignancy in patients with residual masses greater than 1 cm. Lymphovascular invasion is associated with the presence of occult metastases in patients with clinical stage I NSGCT and has no impact on the histology of residual masses after chemotherapy.

14. **d. Observation. Patients at low risk for metastatic disease with indeterminate CT findings should be closely observed because small (<1 cm) retroperitoneal lesions may represent false-positive findings, particularly if they are located outside the primary landing zone.** A CT-guided biopsy would be technically difficult to perform, given the lesion size and its proximity to the renal vessels.
15. **b. Lymphovascular invasion. Lymphovascular invasion is associated with the presence of occult metastases in patients with clinical stage I NSGCT, but it has not been associated with an increased risk of distant metastases in patients with clinical or pathologic stage II disease.** Elevated postorchiectomy serum tumor markers, bulky (>3 cm) retroperitoneal masses, or retroperitoneal lymphadenopathy outside the primary landing zone are associated with an increased risk of systemic relapse after RPLND. Thus, clinical stage IIA-B patients with these features are recommended to receive induction chemotherapy. Scrotal invasion by the primary tumor is associated with an increased risk of metastasis to the inguinal lymph nodes, which are considered nonregional lymph nodes.
16. **d. Poor-risk disease at diagnosis by IGCCCG criteria.** Although patients with poor-risk GCT have diminished survival and are more likely to have viable malignancy or incomplete resection at postchemotherapy RPLND, IGCCCG risk category is not a predictor of relapse independent of the histology of resected masses or completeness of resection. “Desperation” postchemotherapy RPLND in the setting of rising serum tumor markers after second- or third-line chemotherapy and reoperative RPLND are other conditions associated with an increased risk of relapse.

17. **c. Bilateral postchemotherapy RPLND. Approximately one third of patients will have residual masses at multiple anatomic sites, and these patients should undergo resection of all sites of measurable residual disease because discordant histology between anatomic sites is reported in 22% to 46% of cases. However, the presence of necrosis in postchemotherapy RPLND specimens is highly predictive of necrosis at other sites.** Thus, postchemotherapy RPLND should be performed before resection of residual masses at other sites. Observation of small residual masses at other sites is a reasonable option if the histology of the RPLND specimen is necrosis. Patients with viable malignancy discovered at postchemotherapy resection should have all residual masses resected and are usually treated with an additional two cycles of chemotherapy. A full course of second-line chemotherapy is reserved for patients with either serologic or radiographic progression during or after first-line chemotherapy.
18. **e. The outcome is poor relative to those with early NSGCT relapse.** Until recently, late relapse has been associated with a worse prognosis than early relapse, although more recent data suggest these patient groups have a similar probability of cure. Disease-free rates of 50% to 60% are reported after treatment of early and late relapse.
19. **b. CT-guided biopsy of the para-aortic mass. Patients with good-risk metastatic NSGCT who have dramatic progression of their disease with first-line chemotherapy despite normalization of serum tumor marker levels should be considered to have either growing teratoma syndrome or teratoma with malignant transformation.** The presence of an enlarging solid mass and new sites of disease suggest a malignant process. An enlarging mass only with cystic appearance is more suggestive of growing teratoma syndrome. A CT-guided biopsy to identify the presence of malignant transformation is indicated because this finding may influence the choice of chemotherapy.
20. **e. Similar efficacy to cisplatin.** All of the randomized trials in advanced GCT in which a cisplatin-based regimen has been compared with a carboplatin-based regimen have reported superior

outcomes with cisplatin. The rationale for single-agent carboplatin is based on reduced toxicity compared with cisplatin and 65% to 90% response rates reported in studies of carboplatin in advanced seminoma.

21. **d. Ejaculatory dysfunction.** Dog-leg radiotherapy for clinical stage I seminoma is associated with infertility due to the direct effects of radiation on the germinal epithelium with resultant impaired spermatogenesis. Infertility related to ejaculatory dysfunction is not associated with radiation therapy and is most commonly associated with RPLND.
22. **a. Utility of serum tumor markers to identify relapse at an early and curable stage.** Only 15% of seminomas produce elevations in serum hCG, and serum tumor marker levels are uncommonly elevated in most patients with clinical stage I seminoma at diagnosis or at the time of relapse. This contrasts to clinical stage I NGSCT, in which serum tumor markers are commonly the first (and only) manifestation of disease relapse.
23. **c. Observation. In contrast to advanced NSGCT, only 10% of residual masses in advanced seminoma after first-line chemotherapy contain viable malignancy (90% contain fibrosis/necrosis only), and residual teratoma is less of a consideration.** Spontaneous resolution of these masses will occur in most cases. **Approximately 30% of discrete residual masses greater than 3 cm will contain viable malignancy. FDG-PET is a useful adjunct to postchemotherapy staging CT to determine the need for postchemotherapy surgical resection. Residual masses larger than 3 cm that are PET negative and those less than 3 cm can be safely observed because of the high probability of necrosis/fibrosis. FDG-PET has no role in the characterization of residual masses less than 3 cm.**
24. **d. CT of the head. Choriocarcinoma spreads hematogenously and widely. Brain metastases should be suspected in any patient with a very high hCG level.** Thus, patients with high hCG levels at diagnosis should have staging CT or magnetic resonance imaging (MRI) studies of the brain. **Choriocarcinomas are highly vascular and tend to hemorrhage during chemotherapy, which**

may have catastrophic consequences in those patients with brain metastases. Brain metastases are also associated with a poor prognosis, and these patients should receive four cycles of bleomycin-etoposide-cisplatin as first-line chemotherapy, as should any patient with an hCG level over 5000 mU/mL at the time chemotherapy is initiated.

25. **a. Two cycles of platin-based chemotherapy do not increase one's risk of developing cardiovascular disease or secondary malignant neoplasm (SMN).** Although the risk of late complications of chemotherapy is dose dependent, there appears to be no safe lower limit. Thus, even patients receiving one to two cycles of platin-based chemotherapy may have an increased risk of late toxicity.
26. **b. i and iii.** Unlike GCT, Leydig cell tumors are not associated with a history of cryptorchidism, and bilateral tumors have not been reported.
27. **a. Inguinal orchiectomy followed by adjuvant radiotherapy.** A large infiltrative mass involving the spermatic cord in an adult man is a sarcoma until proved otherwise. **The low-intensity signal on CT and patient age make liposarcoma the most common histology. Paratesticular liposarcomas rarely metastasize but tend to recur locally. Thus, adjuvant radiotherapy may be used to decrease the risk of local recurrence.**

Pathology

1. **b. Has a significant chance of developing a germ cell tumor in the left testis.** The figure illustrates germ cell neoplasia in situ as evidenced by enlarged hyperchromatic nuclei and a lack of a spermatogenesis. **This carries a 50% risk of developing a germ cell tumor.**
2. **d. Be advised to have radiation therapy to the retroperitoneum.** Notice in the figure the sheathlike pattern of small cells interspersed with fibrous septa that contain lymphocytes, the hallmarks of seminoma. Although observation is an option, most

would recommend radiation to the retroperitoneum, because seminoma is very sensitive to radiation and the morbidity is low—although there is a risk for the development of secondary malignancies over the long term.

3. **a. Follow markers and check half-life.** The figure demonstrates seminoma with syncytiotrophoblasts. Approximately 15% of patients with seminoma have elevated hCG and will demonstrate syncytiotrophoblasts. Following orchiectomy, the hCG should decline according to its 24-hour half-life. This should be determined first before any treatment decisions are made.
4. **d. Not have any treatment.** The figure shows a spermatocytic tumor that has a very low malignant potential. Notice the small basophilic cells and the multinucleated tumor giant cell, which are characteristic for spermatocytic tumor.
5. **a. Have induction chemotherapy.** This patient has an embryonal carcinoma: notice the primitive, anaplastic epithelial cells. With persistently elevated serum markers, the patient should undergo induction chemotherapy. Surgery is not indicated, radiation therapy is inappropriate, and there is no reason to delay.
6. **a. Have an retroperitoneal lymphadenectomy dissection (RPLND).** The tumor depicted is a teratoma: notice the mature enteric epithelium. Because the specimen is in the primary tumor, there is a high likelihood that there is residual teratoma in the retroperitoneal mass. This tumor is chemoinsensitive and should be resected.

Imaging

1. **b. Testicular neoplasm.** The ultrasound image shows an irregular vascular mass in the left testis that also has microlithiasis. This is most consistent with a testicular neoplasm. It is not unusual for patients to have a history of groin trauma before presentation.
2. **c. Chemotherapy.** The CT image shows a large (>5 cm) para-aortic mass that represents metastatic adenopathy. Because this represents

bulky retroperitoneal disease (stage IIC), chemotherapy is the best option.

Chapter Review

1. GCTs occur bilaterally approximately 2% of the time. The risk factors for developing GCTs include cryptorchidism, a family history of testicular cancer, a previous history of testicular cancer, and GCNIS.
2. In men with a history of GCTs, the finding of testicular microlithiasis on ultrasonography in the contralateral testis is associated with an increased risk of intratubular germ cell neoplasia; the significance of microlithiasis in the general population, however, is unclear.
3. One percent to five percent of GCTs are extragonadal; they are generally less sensitive to chemotherapy and are more likely to contain yolk sac tumor elements than tumors arising in the testis.
4. On rare occasion, teratomas may transform into somatic malignancies, such as rhabdomyosarcoma, adenocarcinoma, or neuroendocrine tumors.
5. Two-thirds of patients with GCTs have diminished fertility.
6. Choriocarcinomas and seminomas do not produce AFP.
7. The half-life of AFP is 5 to 7 days, hCG is 24 to 36 hours, and LDH is 24 hours.
8. The primary landing zone in the retroperitoneum for right testicular tumors is the interaortocaval lymph nodes; for left testicular tumors, it is the periaortic lymph nodes; the pattern of lymph drainage in the retroperitoneum is from right to left.
9. Patients with persistently elevated AFP and hCG after orchiectomy are given induction chemotherapy.
10. In clinical stage I disease, approximately 25% of patients will have metastases.
11. Lymphovascular invasion and a prominent component of embryonal carcinoma are risk factors for metastases in NSGCTs.
12. In seminomas, risk factors for metastases are rete testis involvement and tumor size greater than 4 cm.

13. Patients with bulky retroperitoneal lymph node disease greater than 3 cm should receive induction chemotherapy.
14. After initial treatment, patients with enlargement of a retroperitoneal mass or an increase in markers should undergo salvage chemotherapy. Consideration may be given to a CT-guided biopsy under selected circumstances.
15. Patients with an NSGCT, undetectable markers, and a residual mass greater than 1 cm after chemotherapy should undergo surgical resection.
16. Approximately half of those patients who have surgical resection of a retroperitoneal mass following chemotherapy will harbor teratoma or a viable malignancy. The remainder will have fibrosis.
17. Patients with viable malignancy in residual masses after salvage chemotherapy have a poor prognosis.
18. Predictors of relapse in patients with stage I seminoma on surveillance include rete testis invasion and size of tumor greater than 4 cm. Lymphovascular invasion is not predictive as it is in NSGCT.
19. In patients with seminomas who are treated with chemotherapy, the size of the residual mass is highly predictive of viable tumor. Masses less than 3 cm rarely have viable tumor in them, whereas about a third of residual masses greater than 3 cm contain viable malignancy. FDG-PET is a useful adjunct to postchemotherapy staging CT to determine the need for postchemotherapy surgical resection. Residual masses larger than 3 cm that are PET negative and those less than 3 cm can be safely observed because of the high probability of necrosis/fibrosis.
20. Late toxicity of chemotherapy includes peripheral neuropathy, Raynaud phenomenon, hearing loss, hypogonadism and infertility, secondary malignant neoplasms, and cardiovascular disease.
21. There is an increased number of copies of genetic material from the short arm of chromosome 12 in germ cell tumors.
22. There is no clinical distinction between immature and mature teratoma. Teratomas are resistant to chemotherapy. They also tend to be infiltrative when large and can be extremely difficult to resect.

23. Of patients with testicular tumors, 52% are oligospermic and 10% are azoospermic at presentation.
24. Of patients who receive radiation as treatment for intratubular germ cell neoplasia, 40% require testosterone supplementation.
25. The risk for a secondary malignancy after radiation therapy for seminoma is 18% at 25 years.
26. Ninety percent of Leydig cell tumors and Sertoli cell tumors are benign and 10% are malignant.
27. The most common testicular neoplasm in men older than 50 years is lymphoma.
28. Cystadenoma of the epididymis is associated with von Hippel-Lindau syndrome; adenomatoid tumor of the epididymis is benign.
29. Liposarcoma is the most common paratesticular tumor in the adult. Rhabdomyosarcoma is the most common paratesticular tumor in the child.
30. GCNIS is the common precursor lesion for all types of adult male GCT, with the exception of spermatocytic tumor.
31. Choriocarcinoma has a propensity for hematogenous dissemination. Yolk sac tumors in children are thought to spread hematogenously as well.
32. Pure embryonal carcinoma may produce both AFP and hCG. Pure seminoma is associated with elevated serum hCG levels in 15% of cases but does not produce AFP. Pure teratoma typically is not associated with elevated serum tumor markers, although slightly elevated AFP levels may be observed. Choriocarcinoma is uniformly associated with elevated hCG levels but does not produce AFP. Most yolk sac tumors produce AFP, but they do not produce hCG.
33. Testis-sparing surgery should be considered only in patients with suspected GCT who have normal testicular androgen production and who have a small (<2 cm) tumor either in a solitary testis or in the setting of bilateral synchronous testicular GCT. Testis-sparing surgery should not be performed in patients with suspected GCT who have a normal contralateral testis. Testis-sparing surgery may also be considered in patients with suspected benign testicular lesions such as an epidermoid cyst or adenomatoid tumor arising from the tunica albuginea.

34. Absence of teratoma in the primary tumor, prechemotherapy and postchemotherapy mass size, and percentage shrinkage of mass with chemotherapy are all associated with the presence of necrosis/fibrosis in residual masses after first-line chemotherapy. However, none of these factors (alone or together) is sufficiently accurate to exclude the presence of residual teratoma or viable malignancy in patients with residual masses greater than 1 cm.
35. Approximately one-third of patients who have residual masses following chemotherapy will have residual masses at multiple anatomic sites (sites outside the retroperitoneum), and these patients should undergo resection of all sites of measurable residual disease because discordant histology between anatomic sites is reported in 22% to 46% of cases. However, the presence of necrosis in postchemotherapy RPLND specimens is highly predictive of necrosis at other sites outside the retroperitoneum.

77

Surgery of Testicular Tumors

Stephen Riggs, Kris E. Gaston, and Peter E. Clark

Keywords

orchiectomy; retroperitoneal lymph node dissection; surgical technique; testis cancer

Questions

1. Which of the following nerves is at risk for injury during radical orchiectomy?
    a. Genitofemoral nerve
    b. Ilioinguinal nerve
    c. Obturator nerve
    d. Lateral femoral cutaneous nerve
    e. Pudendal nerve
2. Which of the following may be an indication to consider partial orchiectomy for a patient with a testicular mass?
    a. A polar tumor less than 2 cm in greatest dimension
    b. A normal contralateral testicle
    c. Hypogonadism
    d. Suspicion for benign tumor
    e. Infertility
3. Critical elements to clinically staging testicular cancer routinely include all of the following EXCEPT:
    a. radical orchiectomy

b. chest radiograph
c. whole-body positron emission tomography (PET) scan
d. serum α-fetoprotein (AFP), human chorionic gonadotropin (hCG), and lactate dehydrogenase (LDH)
e. contrasted computed tomography (CT) scan of the abdomen and pelvis

4. The incidence of perioperative acute respiratory distress syndrome (ARDS) in patients with prior receipt of bleomycin can be minimized by all of the following EXCEPT:
a. avoidance of the Trendelenburg position
b. keeping the FiO_2 as low as possible
c. minimization of intraoperative fluid resuscitation
d. minimization of postoperative fluid resuscitation
e. three cycles of Bleomycin rather than four

5. Performing the aortic split-and-roll before that of the inferior vena cava (IVC):
a. allows prospective identification of right accessory lower pole renal arteries not identified on preoperative imaging.
b. facilitates identification of right-sided postganglionic sympathetic nerves as they cross over the aorta.
c. minimizes risk of left ureteral injury.
d. increases risk if injury to the inferior mesenteric artery.
e. should never be performed.

6. The ureter is typically located:
a. anterior to the ipsilateral renal artery.
b. anterior to the ipsilateral retroperitoneal nodal packet.
c. posterior to the ipsilateral gonadal vein adjacent to the lower pole of the ipsilateral kidney.
d. anterior to the ipsilateral gonadal vein adjacent to the lower pole of the ipsilateral kidney.
e. posterior to the ipsilateral common iliac artery.

7. Which of the following anatomic structures demonstrates the most predictable and constant anatomy?
a. Postganglionic sympathetic nerve fibers
b. Lumbar arteries

c. Lumbar veins
d. Number of nodes in each retroperitoneal packet
e. Lymphatic vessels

8. The cisterna chylae is located:
   a. immediately posterolateral to the IVC, just cephalad to the right renal artery.
   b. immediately posterolateral to the IVC, just inferior to the right renal artery.
   c. immediately posterolateral to the aorta, just cephalad to the left renal artery.
   d. immediately posterolateral to the aorta, just inferior to the left renal artery.
   e. immediately posteromedial to the aorta, just cephalad to the right renal artery.
9. The most common auxiliary procedure required to ensure complete resection of residual tumor at postchemotherapy retroperitoneal lymph node dissection (PC)-RPLND is:
   a. IVC resection.
   b. retrocrural resection.
   c. nephrectomy.
   d. pelvic resection.
   e. aortic resection.
10. All of the following are associated with an increased risk of nephrectomy EXCEPT:
   a. Left-sided primary testicular tumor
   b. Prior receipt of salvage chemotherapy
   c. Larger retroperitoneal mass size
   d. Presence of ipsilateral accessory lower pole renal arteries
   e. Elevated serum tumor markers (STMs) at PC-RPLND
11. The histology encountered most often at resection of residual hepatic lesions after chemotherapy is:
   a. viable malignancy.
   b. teratoma.
   c. fibrosis/necrosis.
   d. somatic-type malignancy.
   e. hemangioma.

12. All of the following are associated with an increased risk of pelvic germ cell tumor (GCT) metastases EXCEPT:
    a. prior groin surgery.
    b. late relapse.
    c. require repeat RPLND.
    d. congenital absence of the vasa deferentia.
    e. prior chemotherapy.
13. All of the following are patient selection criteria for a unilateral modified template PC-RPLND (rather than full bilateral template) EXCEPT:
    a. normal STMs.
    b. International Germ Cell Cancer Collaborative Group (IGCCCG), good risk only.
    c. IGCCCG good or intermediate risk.
    d. residual mass less than 5 cm.
    e. well-defined mass confined to the primary landing zone before and after chemotherapy.
14. With regard to the use of postoperative adjuvant cisplatin-based chemotherapy in patients demonstrating pathologic stage IIA-B disease at primary RPLND, all of the following are true EXCEPT:
    a. it spares one to two cycles of chemotherapy for those patients destined to recur on postoperative observation.
    b. it nearly eliminates postoperative recurrences.
    c. it improves overall and cancer-specific survival.
    d. it results in overtreatment of 50% to 70% of patients if given to all patients.
    e. it is typically given in two cycles.
15. Which of the following characteristics has been associated with increased recurrence rate when teratoma is encountered at PC-RPLND?
    a. Extranodal extension
    b. Presence of somatic type malignancy
    c. Right-sided primary GCT
    d. Presence of immature teratoma
    e. Number of lymph nodes removed

16. All of the following factors have been associated with worse prognosis when viable GCT is encountered at PC-RPLND EXCEPT:
    a. incomplete resection.
    b. less than 10% viable GCT in resection specimen.
    c. IGCCCG intermediate or poor risk status.
    d. prior receipt of salvage chemotherapy.
    e. immature teratoma in the specimen.
17. Which of the following criteria is an accepted indication for two cycles of adjuvant chemotherapy after primary RPLND?
    a. Number of positive nodes relative to the number removed
    b. Teratoma only nodal metastases
    c. Extranodal extension of GCT
    d. pN1 disease or higher
    e. pN2 disease or higher
18. A patient has an isolated resectable residual retroperitoneal mass after induction chemotherapy without radiographic evidence of disease outside the retroperitoneum, but tumor markers have failed to normalize. All of the following are reasonable indications for the consideration of desperation RPLND EXCEPT:
    a. declining AFP after induction chemotherapy.
    b. slowly rising AFP after a complete serologic response to induction chemotherapy.
    c. all potentially curative chemotherapeutic options have been exhausted.
    d. persistently rising STMs through induction chemotherapy.
    e. plateauing AFP after induction chemotherapy.
19. Reoperative RPLND is thought to indicate a technical failure at prior RPLND. All of the following findings supportive of this hypothesis have been reported in the literature EXCEPT:
    a. the primary landing zone is the most common site of retroperitoneal recurrence.
    b. ipsilateral pelvic recurrences are common.
    c. incomplete ipsilateral lumbar vessel ligation encountered at reoperative RPLND has been associated with ipsilateral infield recurrence.

d. unresected ipsilateral gonadal vessels are frequently encountered at reoperative RPLND.
e. the retroaortic and retrocaval regions are frequent sites of recurrence.

20. All of the following are true regarding late relapse of GCT EXCEPT:
    a. yolk sac tumor is the most common viable histology encountered.
    b. first-line treatment is generally systemic chemotherapy followed by consolidative PC-RPLND.
    c. patients who are chemotherapy-naïve demonstrate superior survival outcomes.
    d. GCT with somatic-type malignancy is seen with increased frequency in this population.
    e. the retroperitoneum is the most common site of late relapse.
21. What percentage of patients presenting with GCT have abnormal parameters on semen analysis?
    a. Less than 10%
    b. 20% to 60%
    c. 70% to 80%
    d. Greater than 90%
22. The processes required to ensure antegrade ejaculation of sperm-containing semen include all of the following EXCEPT:
    a. seminal emission through vasa deferentia.
    b. closure of the bladder neck.
    c. smooth muscle contraction of the prostate.
    d. penile erection.
    e. input from sympathetic fibers arising from L1 to L4.
23. All of the following interventions have demonstrated efficacy in managing chylous ascites EXCEPT:
    a. medium-chain triglyceride (MCT) diet.
    b. total parenteral nutrition.
    c. subcutaneous octreotide.
    d. limiting fat intake preoperatively.
    e. placement of peritoneovenous shunt.

24. Which of the following patients is at the greatest risk for neurologic compromise due to spinal ischemia?
    a. 32-year-old male undergoing resection of left para-aortic mass with apparent aortic invasion who will most likely require resection of the infrahilar aorta and tube graft reconstruction
    b. 29-year-old male with large-volume left para-aortic and interaortocaval masses that extend through the retrocrural region into the middle visceral mediastinum
    c. 27-year-old male with a completed occluded IVC due to large interaortocaval, retrocaval, and right paracaval masses with a tumor thrombus up the inferior border of the right renal vein
    d. 31-year-old male with a large infrarenal left para-aortic mass that is found to be invading the L2 vertebral foramina during resection

Answers

1. **b. Ilioinguinal nerve.** The ilioinguinal nerve will be encountered immediately on opening the external oblique fascia and entering the inguinal canal. It courses parallel to the spermatic cord along the cephalad aspect of its anterior surface. Care should be taken to preserve this structure in order to prevent postoperative numbness and paresthesia of the ipsilateral medial thigh and scrotum.
2. **d. Suspicion for benign tumor.** For obvious reasons, patients who are suspected to have a benign tumor are prime candidates for partial orchiectomy. Patients with larger tumors in the setting of normal contralateral testicle should be managed with radical orchiectomy as they will not likely suffer permanent postoperative infertility or hypogonadism. In addition, the benefits of partial orchiectomy with the goal of saving poorly functioning testicular parenchyma in a patient with baseline infertility and/or hypogonadism are most likely outweighed by the potential for incomplete resection of tumor and/or ipsilateral testicular recurrence.

3. **c. Whole-body positron emission tomography (PET) scan.** Testicular GCT patients are clinically staged with radical orchiectomy (T stage), chest imaging such as a chest X-ray or Chest CT scan, a CT scan or magnetic resonance imaging (MRI) of the abdomen and pelvis (N and M stages), and postorchiectomy STMs (S stage). PET scans have no role in the initial clinical staging of seminomas or nonseminomas. **PET scans do have a role in evaluating postchemotherapy residual masses in cases of pure seminoma that are over 3 cm in size.**
4. **a. Avoidance of the Trendelenburg position.** Postoperative ARDS in patients with prior receipt of bleomycin is rare but most commonly encountered in patients having received four courses of bleomycin (rather than three) with significant pulmonary tumor burden. The two intraoperative/perioperative factors that have been associated with an increased risk of developing postoperative ARDS have been (1) exposure to high FiO_2 and (2) high-volume administration of intravenous fluids and blood products. Anesthesiologists should be made aware of these risk factors.
5. **a. Allows prospective identification of right accessory lower pole renal arteries not identified on preoperative imaging.** The advantage of splitting of the aorta before the IVC is that it allows for the prospective identification of right accessory lower pole renal arteries not identified on preoperative imaging. Such vessels may be inadvertently divided if the IVC split is performed first. This can lead to significant blood loss and renal parenchymal loss. **The disadvantage of splitting of the aorta first is inadvertent division of postganglionic sympathetic fibers involved in ejaculation. This can be minimized by stopping the aortic split at the inferior mesenteric artery and prospectively identifying the sympathetic efferents before continuing the split caudally.**
6. **c. Posterior to the ipsilateral gonadal vein adjacent to the lower pole of the ipsilateral kidney. The ureters provide the anatomic landmark for the lateral border of dissection for the para-aortic and paracaval lymph node packets.** However, the ureters are vulnerable to injury if not prospectively identified. The ureter

can be easily found at the inferior border of the ipsilateral kidney where it typically passes behind the ipsilateral gonadal vein. The ureter passes anterior to the ipsilateral common iliac artery as it descends into the pelvis.

7. **b. Lumbar arteries. There are three paired lumbar arteries located between the renal hilum and the aortic bifurcation in nearly all patients.** The only subtle anomaly commonly encountered with aortic lumbars is that the paired lumbars may exit the aorta through a short common trunk before bifurcating, a variant that can result in significant blood loss if not identified before these structures are ligated and divided. Lumbar vein anatomy is high variable, with these structures varying widely in location and caliber. While the number of postganglionic sympathetic roots in the field of RPLND tends to be relatively constant at four per side, the exact locations, paths, and patterns of fusion to adjacent roots is highly variable. Studies on lymph node counts have universally reported high standards of deviation and interquartile ranges.
8. **e. Immediately posteromedial to the aorta just cephalad to the right renal artery.** Failure to meticulously ligate large-caliber lymphatics in the region of the cisterna chylae can predispose patients to troublesome postoperative chylous ascites. Thus the use of clips and/or ties at the superior extent of the interaortocaval and para-aortic packets, as well as during retrocrural dissections, is advised.
9. **c. Nephrectomy.** The incidence of auxiliary procedures at the time of PC-RPLND ranges between 23% and 45% in the literature. The most common auxiliary procedure is nephrectomy followed by vascular interventions. The larger the volume of the residual retroperitoneal mass, the greater the need for possible auxiliary procedures.
10. **d. Presence of ipsilateral accessory lower pole renal arteries.** Reported nephrectomy rates at PC-RPLND have ranged from 5% to 31%. Increased nephrectomy rates have been reported for patients undergoing salvage RPLND, desperation RPLND, reoperative RPLND, and resection of late relapse. In addition, large

retroperitoneal mass size and left-sided primary tumors are associated with increased rates of nephrectomy.

11. **c. Fibrosis/necrosis.** Approximately three quarters of residual hepatic lesions after chemotherapy will demonstrate fibrosis/necrosis only at resection. There is a 94% concordance between retroperitoneal and hepatic fibrosis/necrosis. Thus observation should be considered in patients with retroperitoneal necrosis and residual hepatic lesions that would require a potentially morbid resection.
12. **d. Congenital absence of the vasa deferentia. Although the pelvis is the caudal extension of the retroperitoneum, it is not a common site of GCT metastases.** Pelvic disease in metastatic testes cancer is rarely observed outside of prior groin surgery, repeat RPLND, or late relapsed disease. Data from Indiana University from 1990 to 2009 identified 2722 patients who underwent RPLND, of which 134 had pelvic disease—14% had prior groin surgery, 98% had prior chemotherapy, 19.4% underwent prior RPLND, and 24% had late stage relapse.
13. **b. International Germ Cell Cancer Collaborative Group(IGCCCG), good risk only.** The application of modified unilateral templates in the setting of PC-RPLND is critically reliant on proper patient selection. While no universal consensus exists, based on the reports to date patients who are potential candidates for modified template PC-RPLND should probably meet the following criteria. **(**1) Well-defined lesion measuring 5 cm or less confined to the primary landing zone of the primary tumor on imaging before and after chemotherapy (see Fig. 77.1 for representative CT examples). (2) Normal STMs. (3) IGCCCG good or intermediate risk. Following selection criteria similar to these has been reported in PC-RPLND with excellent local control, including in-field retroperitoneal recurrence rates of 0% to 1% and cancer specific survival rates of 98% to 100% at a follow-up of 2.6 to 10.4 years, while maintaining antegrade ejaculation in over 85% of men.
14. **c. Improves overall and cancer-specific survival.** Adjuvant chemotherapy for pathologic stage IIA-B disease encountered at

primary RPLND consists of two cycles of bleomycin, etoposide, and cisplatin (BEP) or two cycles of etoposide and cisplatin (EP). **This nearly eliminates the risk of recurrence but has no effect on cancer-specific survival, which approaches 100% whether chemotherapy is administered in the adjuvant setting or reserved for patients who experience relapse.** Patients who relapse will require full induction chemotherapy consisting of BEP×3 or EP×4. When pathologic stage IIA-B patients are observed after primary RPLND, approximately 30% to 50% of patients relapse.

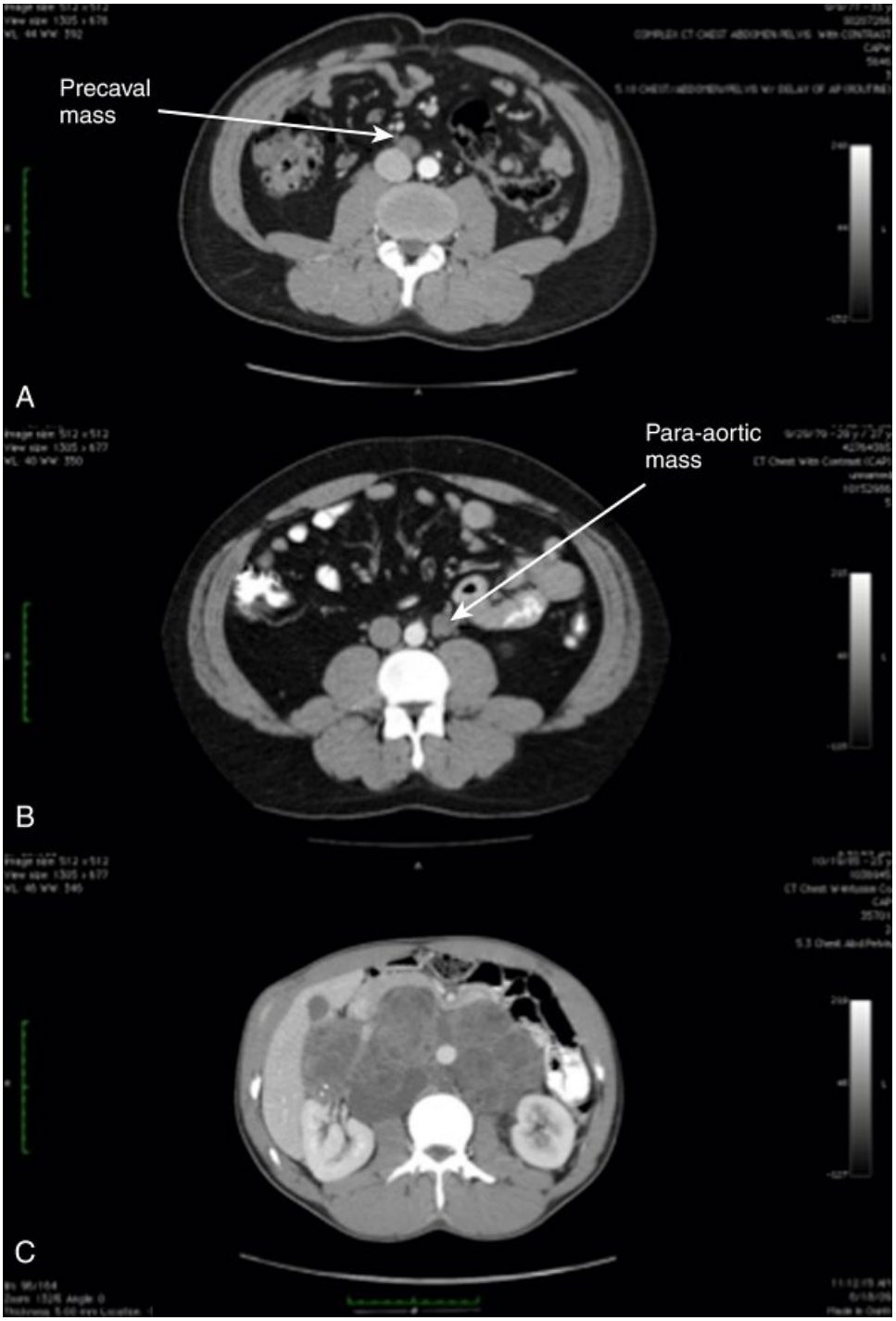


FIG. 77.1 Computed tomography images of postchemotherapy residual retroperitoneal masses. (A) This patient could be considered a candidate for modified right template postchemotherapy retroperitoneal lymph node dissection (PC-RPLND). (B) This patient could be considered a candidate for modified left template PC-RPLND. (C) This patient would require an extensive bilateral PC-RPLND.

15. **b. Presence of somatic type malignancy.** Patients with somatic-type malignancy and mediastinal primary GCT disease have a higher risk of postoperative recurrence. **Although once thought to be a marker of more aggressive behavior, immature teratoma has demonstrated no decrement in survival outcomes compared with mature teratoma.** Laterality, number of nodes removed, and extranodal extension have not been shown to have any relationship to the risk of relapse in the setting of teratoma only residual disease after PC-RPLND.
16. **e. Immature teratoma in the specimen.** In a large multicenter review by Fizazi and colleagues, incomplete resection, proportion of viable GCT in resection specimen, and IGCCCG risk status were used to classify patients into three risk categories that predicted survival outcomes. When evaluating patients with viable GCT at PC-RPLND, prior receipt of salvage chemotherapy has consistently been associated with significantly poorer survival outcomes. Although once thought to be a marker of more aggressive behavior, immature teratoma has demonstrated no decrement in survival outcomes.
17. **e. pN2 disease or higher. There is relatively broad consensus that many patients with pN1 disease at primary RPLND who are reliable can be safely managed with observation.** The management of pN2 disease is more controversial with some centers advocating two cycles of adjuvant chemotherapy while others would manage many of these patients with observation. There have been a variety of other histologic variables tested for their prognostic ability to predict disease recurrence. These include such parameters as the number of positive nodes as a proportion of the number removed, the histologic type of GCT in the metastases, and the presence or absence of extranodal extension. **None of these have proven reliable enough that they can accurately predict those who are at high risk for recurrence and warrant up front adjuvant therapy. The one exception is that the finding of only teratoma in the retroperitoneum is associated with very low relapse rates, and therefore observation is warranted.**

18. **d. Persistently rising STMs through induction chemotherapy.** Investigators at Indiana University reported a 53.9% cancer-specific survival rate at a median follow-up of 6 years when the following criteria outlined were used to select patients to undergo desperation PC-RPLND. (1) Declining or Plateauing AFP after induction chemotherapy, a slowly rising AFP after a complete serologic response to induction chemotherapy, and all potentially curative chemotherapeutic options have been exhausted. A patient with persistently rising STMs during chemotherapy should receive either standard or high-dose salvage chemotherapy.
19. **b. Ipsilateral pelvic recurrences are common.** Reoperative RPLND indicates a technical failure at prior RPLND in the majority of cases. Several reviews of reoperative experiences at high-volume centers have revealed that primary landing zone recurrence as well as signs of an inadequate prior resection (incomplete/absent lumbar vessel ligation, unresected ipsilateral gonadal vein, and recurrence posterior to the great vessels) are common in patients experiencing retroperitoneal recurrence after RPLND. Since pelvic lymph nodes are not routinely removed as part of a standard template RPLND disease relapse in this location does not necessarily reflect a technical failure of the any preceding RPLND.
20. **b. First-line treatment is generally systemic chemotherapy followed by consolidative PC-RPLND. Late relapse in patients with prior receipt of chemotherapy is often composed of yolk sac tumor and tends to be relatively chemorefractory regardless of histology. Thus primary management of resectable disease at late relapse is surgical extirpation.** Late relapse in chemotherapy-naïve patients has been associated with improved outcomes likely due to increased susceptibility to chemotherapy.
21. **b. 20% to 60%.** Infertility and subfertility are frequently present at GCT diagnosis. This can present challenges to pretreatment sperm banking. For unclear reasons, parameters occasionally improve after orchiectomy. However, this baseline infertility needs to be

taken into account when evaluating fertility outcomes after treatment of GCT.

22. **d. Penile erection.** Antegrade ejaculation is reliant on **the smooth muscle contraction of the vas deferentia, seminal vesicles, and prostate that results in seminal emission and prostate glandular secretion, as well as the subsequent closure of the bladder neck preventing retrograde ejaculation, are critically reliant on input from the postganglionic sympathetic fibers that emanate from L1 to L4.** Penile erection is necessary for vaginal penetration, but not ejaculation. Notably, erection is a parasympathetic process.
23. **d. Limiting fat intake preoperatively.** Although it is a troublesome postoperative complication, chylous ascites tends to be transient, and management is usually directed at draining accumulated fluid and minimizing further production by instituting an MCT diet, progressing to total parenteral nutrition if the MCT diet fails, and possibly starting subcutaneous octreotide. A peritoneovenous shunt is an intervention of last resort. Limiting preoperative fat intake has not been shown to decrease the incidence of chylous ascites.
24. **b. 29-year-old male with large-volume left para-aortic and interaortocaval masses that extend through the retrocrural region into the middle visceral mediastinum.** Spinal ischemia leading to neurologic compromise is extremely rare. Patients at risk for this potentially devastating complication are those who will require resection of more than three sequential lumbar artery pairs. This tends to be in patients with high-volume para-aortic disease in the retroperitoneum, retrocrural region, and visceral mediastinum.

Chapter Review

1. The proper performance of a radical inguinal orchiectomy includes mobilizing the cord 1 to 2 cm proximal to the internal inguinal ring and individually ligating the vas deferens and cord vessels with silk

sutures so that the stump may be identified if an RPLND is performed.

2. When an orchiectomy has been performed through the scrotum in patients who have a low-stage seminoma, the radiation portals should be extended to include the ipsilateral groin and scrotum; for those with low-stage nonseminomatous germ cell tumor (NSGCT), the scrotal scar should be excised along with the spermatic cord remnant; and for those who have received a full cycle of platinum-based chemotherapy, only the cord stump need be removed at the time of RPLND.
3. The right testicular lymphatics drain to the interaortocaval lymph nodes followed by paracaval and pericaval nodes. The left testicular lymphatics drain to the periaortic and preaortic lymph nodes.
4. Contralateral lymphatic flow is more commonly seen in right-sided tumors than left-sided tumors.
5. There is a 20% to 30% incidence of positive nodes in clinical stage I disease and approximately a 25% relapse rate in such patients who are placed on a surveillance protocol.
6. Suprahilar metastases are rare in low-stage NSGCT. Three percent to 23% of patients with positive nodes at RPLND will have extra-template disease. The most common site of residual suprahilar disease is in the retrocrural space.
7. It is extremely important, when performing a primary RPLND, to secure all lymphatic vessels with either clips or ties, particularly in the region of the right renal artery and diaphragmatic crus, to minimize injury to the cisterna chyli, which could result in chylous ascites.
8. In the presence of documented or suspected metastatic disease, a full bilateral dissection should be performed. In selected cases, preservation of individual nerve fibers may be performed.
9. The anterior split over the vena cava is not likely to damage nerve fibers; however, the anterior split over the aorta risks injury to these fibers.
10. The most important nerves to preserve antegrade ejaculation are those arising from the L1 to L4 ganglia. To preserve nerve fibers, a

dissection on the aorta should only be performed after the nerve fibers have been identified and isolated.

11. There is a sevenfold increase in cardiovascular complications in men treated with platinum-based chemotherapy.
12. All patients should be given the opportunity to bank sperm before RPLND.
13. In clinical stage I, NSGCT lymphovascular invasion; higher T stage; tumor involvement of the cord, capsule, or scrotum; and a high percentage of embryonal carcinoma are associated with an increased incidence of retroperitoneal relapse. Most relapses occur within the first 2 years and are rare after 5 years. The absence of teratoma in the primary tumor does not preclude its presence in the retroperitoneum.
14. Persistently elevated tumor marker levels after orchiectomy require systemic chemotherapy, as the next step is not RPLND.
15. Patients best suited for RPLND are those with clinical stage IIA and low-volume, less than 3-cm ipsilateral disease (stage IIB).
16. Sixty percent of patients with testicular cancer have subnormal pretreatment semen analyses. Sixty-five percent of men on surveillance can impregnate their partner after orchiectomy.
17. Bilateral RPLND is the standard template for patients with pathologic stage II NSGCT.
18. Bleomycin can cause restrictive pulmonary fibrosis with increased collagen deposition and makes the patient highly susceptible to fluid overload.
19. Increased serum concentration of AFP and hCG after primary platinum-based chemotherapy is usually characterized by unresectable viable tumor, and salvage chemotherapy rather than excision of the mass is recommended.
20. After chemotherapy, retroperitoneal masses are composed of necrosis/fibrosis in 40%, teratoma in 45%, and viable GCT in 15%. A residual mass after salvage chemotherapy is composed of viable GCT in 50%, teratoma in 40%, and fibrosis in 10%. The clinical behavior of a teratoma is unpredictable, and complete resection is required.
21. After chemotherapy, RPLND is indicated for NSGCT primary tumors in which the residual mass is larger than 1 cm. RPLND may

be eliminated after chemotherapy if there was no evidence of the disease in the retroperitoneum before chemotherapy.

22. Late relapses occur in 2% to 4% of patients, and more than half of late relapses occur beyond 10 years, emphasizing the need for prolonged follow-up.
23. If the aortic wall is stripped of its adventitia, it should be replaced with a synthetic graft because delayed rupture may occur.
24. Tumor involvement of the superior mesenteric artery, celiac axis, or porta hepatis usually precludes resection. After chemotherapy, resection of residual masses should be accompanied by a complete RPLND. The standard bilateral dissection is the prudent approach.
25. After chemotherapy for seminoma, residual masses very rarely contain teratoma and are extremely difficult technically to remove. Thus a residual mass less than 3 cm should be observed, patients with masses larger than 3 cm should have PET, and if a viable seminoma is noted, then either additional chemotherapy or RPLND is indicated.
26. After chemotherapy, spermatogenesis may take 3 years to return to normal.
27. In the setting of complete or near-complete occlusion, routine reconstruction of the vena cava following resection is not required.
28. RPLND histology is a strong predictor of histology at thoracotomy: if necrosis is all that is found in the retroperitoneum, 89% of the time the chest lesions will be necrotic.
29. The ureters provide the anatomic landmark for the lateral border of dissection for the para-aortic and paracaval lymph node packets.
30. There are three paired lumbar arteries located between the renal hilum and the aortic bifurcation in nearly all patients.
31. Patients having received salvage chemotherapy who demonstrate viable GCT at PC-RPLND, those demonstrating teratoma only at PC-RPLND, and those demonstrating transformation to somatic-type malignancy have not been shown to benefit from adjuvant chemotherapy.
32. Late relapse in patients with prior receipt of chemotherapy is often composed of yolk sac tumor and tends to be relatively

chemorefractory regardless of histology. Thus primary management of resectable disease at late relapse is surgical extirpation.

78

Laparoscopic and Robotic-Assisted Retroperitoneal Lymphadenectomy for Testicular Tumors

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Keywords

chylous ascites; laparoscopic; lymph node dissection; lymphadenectomy; oncology; retroperitoneal; robotic; testes cancer; testicular cancer

Questions

1. A 23-year old man presents after undergoing transscrotal orchiectomy for presumed hydrocele. Pathologic examination reveals embryonal carcinoma with vascular invasion. Serum levels of tumor markers and results of physical examination and computed tomography (CT) of the chest, abdomen, and pelvis were normal. Which of the following approaches is most appropriate?
    a. Observation
    b. Retroperitoneal lymph node dissection (RPLND)
    c. RPLND plus excision of scrotal scar and remnant cord
    d. RPLND plus scrotectomy and inguinal lymph node dissection
    e. RPLND plus scrotal and inguinal radiation
2. Late relapse is a feature most commonly associated with
    a. seminoma.

b. yolk sac tumor.
c. embryonal carcinoma.
d. choriocarcinoma.
e. teratoma.

3. A 25-year-old man with a stage IIC nonseminomatous germ cell tumor (NSGCT) has completed primary platinum-based chemotherapy. Tumor marker levels have normalized according to appropriate half-life, and he has undergone bilateral postchemotherapy RPLND. Final pathologic analysis reveals a focus of yolk sac tumor. Appropriate therapy at this point is
   a. careful observation.
   b. radiation therapy.
   c. two additional cycles of platinum-based chemotherapy.
   d. four additional cycles of platinum-based chemotherapy.
   e. reexploration in 6 weeks.
4. A 20-year-old man with clinical stage I NSGCT undergoes laparoscopic RPLND. During surgery, a 2-cm lymph node is encountered. Which of the following is the most appropriate next step?
   a. Abort the procedure and administer chemotherapy.
   b. Convert to an open procedure.
   c. Perform a unilateral template dissection and administer chemotherapy.
   d. Continue the procedure and perform a full bilateral dissection.
   e. None of the above.
5. The most common cause of open conversion during laparoscopic RPLND is
   a. intraoperative discovery of bulky lymphadenopathy.
   b. failure to progress.
   c. bowel injury.
   d. hypercapnia.
   e. bleeding.
6. Two weeks after laparoscopic RPLND, a patient complains of abdominal distention and emesis. CT reveals ascites. Diagnostic paracentesis confirms the diagnosis of chylous ascites. The next best step is

a. reassurance and discharge.
b. reoperation to identify and treat the source of lymphatic leak.
c. placement of a peritoneal drain and initiation of a low-fat diet.
d. initiation of somatostatin.
e. hydration and initiation of a low-fat diet.

7. A 20-year-old man undergoes laparoscopic RPLND after right radical orchiectomy for an NSGCT. All of the following regions should be dissected clear of all lymphatic tissue EXCEPT which?
   a. Right spermatic cord
   b. Paracaval region
   c. Interaortocaval region
   d. Retrocrural region
   e. Precaval region
8. Potential advantages to laparoscopic compared with open RPLND include all of the following EXCEPT which?
   a. Improved cosmesis
   b. Shorter convalescence
   c. Improved disease-free survival
   d. Shorter interval to chemotherapy when necessary
   e. Faster return to normal activities

Answers

1. **c. RPLND plus excision of scrotal scar and remnant cord.** In the setting of scrotal contamination and clinical stage I disease, the patient is best managed with RPLND and wide excision of the scrotal scar. The remainder of the cord should also be removed. Observation is not optimal because of the presence of vascular invasion and scrotal contamination.
2. **e. Teratoma.** Late relapse of germ cell tumor (GCT) after definitive therapy is defined as recurrence more than 2 years after completion of therapy and occurring without evidence of disease. Teratoma is the most common histologic subtype involved in cases of late relapse. This is likely due to its combination of prolonged doubling time and chemotherapy resistance.

3. **c. Two additional cycles of platinum-based chemotherapy.** The patient's prognosis is related to serum tumor marker level at the time of RPLND, prior treatment burden, and the pathologic findings for the resected specimen. If viable GCT is present at any site but all disease is completely resected, two additional cycles provide survival benefit in this subset of patients. Einhorn reported only 2 long-term survivors of 22 patients (9%) with completely resected viable GCT after cisplatin, bleomycin, and vinblastine chemotherapy if additional postoperative chemotherapy was not given. Fox and colleagues reported that 70% of patients with completely resected viable GCT after primary chemotherapy followed by two cycles of postoperative chemotherapy remained disease free compared with none of 7 patients without additional chemotherapy.
4. **d. Continue the procedure and perform a full bilateral dissection.** Whenever a suspicious lymph node is identified, a full bilateral dissection should be performed. Chemotherapy will not compensate for an inadequate retroperitoneal dissection.
5. **e. Bleeding.** The most common reason for conversion to an open procedure is uncontrollable bleeding, and vascular injury is cited as the most common intraoperative complication. In experienced hands, this occurs less than 5% of the time.
6. **e. Hydration and initiation of a low-fat diet.** Chylous ascites is a devastating complication that occurs less than 2% of the time after primary laparoscopic (and open) RPLND. Initiation of a low-fat diet, hydration, and drainage of the fluid is usually the first step in management. Reoperation is considered only when all other options have been exhausted.
7. **d. Retrocrural region.** The superior boundary of a dissection template does not include lymph nodes superior to the renal hilum.
8. **c. Improved disease-free survival.** No study has demonstrated any difference in disease-free survival comparing open or laparoscopic approaches with node dissection.

Chapter review

1. Patients who have small-volume retroperitoneal disease (pN1) are cured 70% of the time by a radical RPLND.
2. RPLND is the treatment of choice for patients with stage I NSGCTs with teratoma.
3. All patients undergoing RPLND should be offered preoperative sperm banking.
4. In nerve-sparing techniques for RPLND, great care should be taken around the lumbar veins because these are adjacent to the sympathetic chain.

79

Tumors of the Penis

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Keywords

dynamic sentinel node biopsy; human papilloma virus; lymph node metastasis; lymphadenectomy; penectomy; penile cancer; squamous cell carcinoma

Questions

1. Which of the following penile lesions is not typically associated with viral infection?
    a. Balanitis xerotica obliterans
    b. Condylomata acuminatum
    c. Kaposi sarcoma
    d. Bowenoid papulosis
    e. Erythroplasia of Queyrat
2. Which of the following infections is associated with cervical dysplasia?
    a. Human immunodeficiency virus (HIV) infection
    b. Herpesvirus infection
    c. Gonorrhea
    d. Human papillomavirus (HPV) infection
    e. Lymphogranuloma venereum
3. What is the major difference between Bowen disease and erythroplasia of Queyrat?

a. Loss of rete pegs
b. Keratin staining
c. Viral etiologic agents
d. Location
e. Treatment options

4. Kaposi sarcoma of the acquired immunodeficiency syndrome (AIDS)–related (epidemic) type is associated with which of the following etiologic agents?
   a. HPV type 16
   b. Human herpesvirus (HHV) type 8
   c. HPV type 32
   d. *Haemophilus ducreyi* (chancroid [soft chancre])
   e. Coxsackievirus type 23
5. Where do penile cancers most commonly arise?
   a. Glans
   b. Shaft
   c. Frenulum
   d. Coronal sulcus
   e. Scrotum
6. Which of the following is not considered a risk factor for the development of squamous cell carcinoma of the penis?
   a. Cigarette smoke
   b. HPV infection
   c. Phimosis
   d. Gonorrhea
   e. Chewing tobacco
7. All of the following are preventive strategies to decrease the incidence of penile cancer EXCEPT:
   a. circumcision after 21 years of age
   b. HPV vaccination
   c. daily genital hygiene
   d. avoiding tobacco products
   e. circumcision before puberty
8. Which of the following statements regarding penile cancer is FALSE?
   a. Cancer may develop anywhere on the penis.

b. Because of the associated discomfort, patients usually present to physicians within the first month of noting the lesion.
c. Phimosis may obscure the nature of the lesion.
d. Penetration of the Buck fascia and the tunica albuginea by the tumor permits invasion of the vascular corpora.
e. Cancer cells reach the contralateral inguinal region because of lymphatic cross-communications at the base of the penis.

9. Before a treatment plan for penile cancer is initiated, which of the following is TRUE?
    a. Adequate biopsies to determine stage are unimportant because all patients should be treated with amputation.
    b. Radiologic studies play no role in decision making.
    c. Tumor HPV status is critical for determining primary tumor therapy.
    d. Tumor stage, grade, and vascular invasion status all provide prognostically important information.
    e. No disfiguring therapy is indicated, because spontaneous remissions have been noted in approximately 10% of cases.
10. Which of the following statements is TRUE regarding the natural history of penile cancer?
    a. Metastases from the primary tumor often involve lung, liver, or bone as initial sites.
    b. Lymphatic drainage from the primary tumor is ipsilateral alone in most cases.
    c. Metastasis often initially involves spread from the corpora cavernosa to the pelvic lymph nodes.
    d. Metastasis initially involves inguinal lymph nodes beneath the fascia lata.
    e. Metastasis initially involves inguinal lymph nodes above the fascia lata.
11. Which of the following statements concerning hypercalcemia in patients with penile cancer is TRUE?
    a. It is more commonly due to massive bone metastases than bulky soft tissue metastases.
    b. It is often related to uremia due to ureteral obstruction.

c. It may be due to the action of parathyroid hormone-like substances released from the tumor.
d. It is related to the action of osteoblasts on bone formation.
e. It is managed with aggressive diuretic administration as first-line therapy.

12. The following statements are true regarding imaging tests in patients with penile cancer EXCEPT:
    a. Both ultrasonography and MRI lack sensitivity for the detection of corpus cavernosum involvement.
    b. computed tomography (CT) is not an appropriate test for determining primary tumor stage.
    c. CT may be beneficial in detecting enlarged inguinal nodes in obese patients or those who have had prior inguinal therapy.
    d. lymphangiography can detect abnormal architecture in normal-sized lymph nodes.
    e. inguinal palpation is preferred to CT and lymphangiography for determining inguinal nodal status.
13. According to the eighth edition of the American Joint Committee on Cancer Staging System (i.e., tumor, node, metastases [TNM]) for penile cancer, which of the following statements is TRUE?
    a. Primary tumor stage is based on the size of the primary lesion.
    b. Lymph node stage is based in part on the size of an involved node.
    c. Stage T2 tumors invade the corpus spongiosum but not the cavernosum.
    d. Proven pelvic nodal metastases as well as bilateral inguinal metastases are both designated as stage pN3.
    e. Stage T1a tumors involve the dartos fascia and exhibit perineural invasion in less than 10% of the specimen.
14. What is the strongest prognostic factor for survival in penile cancer?
    a. The presence of lymph node metastasis
    b. The grade of the primary tumor
    c. The stage of the primary tumor
    d. Vascular invasion presence in the primary tumor
    e. The extent of lymph node metastasis

15. Criteria for curative surgical resection (>70% 5-year survival) in patients treated for lymph node metastasis include all of the following EXCEPT:
    a. no more than two positive inguinal lymph nodes.
    b. no positive pelvic lymph nodes.
    c. absence of extranodal extension of cancer.
    d. unilateral metastasis.
    e. a single metastasis of only 6 cm.
16. Surgical staging of the inguinal region is strongly considered under all of the following conditions EXCEPT:
    a. palpable adenopathy.
    b. stage T1b or greater primary tumor.
    c. presence of vascular invasion in primary tumor.
    d. presence of predominantly high-grade cancer in primary tumor.
    e. stage Ta tumors.
17. Strategies attempting to minimize the morbidity of inguinal staging in patients with no palpable adenopathy include all the following EXCEPT:
    a. superficial inguinal lymph node dissection.
    b. modified complete inguinal dissection.
    c. standard ilioinguinal dissection.
    d. laparoscopic and robotic inguinal node dissection.
    e. dynamic sentinel node biopsy.
18. Which of the following inguinal staging procedure(s) is/are considered standard for detecting microscopic metastases while limiting both morbidity and false-negative findings?
    a. Sentinel lymph node biopsy
    b. Superficial inguinal dissection
    c. Dynamic sentinel lymph node biopsy
    d. All of the above
    e. b and c only
19. For patients with proven unilateral metastasis involving two or more lymph nodes at presentation, all of the following surgical considerations are true EXCEPT:
    a. ipsilateral ilioinguinal lymphadenectomy should be performed

b. a contralateral staging procedure is not indicated
c. a contralateral staging procedure is indicated
d. both a superficial dissection and a deep ipsilateral dissection are performed
e. ipsilateral pelvic dissection provides useful prognostic information

20. Adjuvant or neoadjuvant chemotherapy should be considered in addition to surgery for all of the following EXCEPT:
    a. single pelvic nodal metastasis
    b. extranodal extension of cancer
    c. fixed inguinal masses
    d. two unilateral inguinal nodes with focal metastases
    e. single 6-cm inguinal lymph node
21. The majority of penile cancers are histologically:
    a. melanoma.
    b. bowenoid papulosis.
    c. squamous cell carcinoma.
    d. epidemic Kaposi sarcoma.
    e. verrucous carcinoma.
22. Which of the following chemotherapeutic agents used in combination therapy for penile cancer has been associated with significant pulmonary toxicity?
    a. Bleomycin
    b. Methotrexate
    c. Cisplatin
    d. 5-Fluorouracil (5-FU)
    e. Paclitaxel
23. Indications for radiation therapy as primary treatment for penile cancer include which of the following?
    a. Young, sexually active patient with a small lesion
    b. Patient refuses surgery
    c. Patient with distant or inoperable metastases who requires local control to the primary tumor
    d. None of the above
    e. a, b, and c
24. Primary penile melanoma is thought to be rare for what reason?

a. Penile skin is protected from exposure to the sun.
b. Keratin content in penile skin is decreased.
c. Penile blood supply precludes such tumor development.
d. Effective topical chemotherapy exists.
e. None of the above.

25. Lymphomatous infiltration of the penis is most likely secondary to which condition?
    a. Autoimmune disorder
    b. Diffuse disease
    c. Metastasis from a distant primary tumor
    d. Chronic infection
    e. Previous venereal infection
26. What is the most frequently encountered sign of metastatic involvement of the penis?
    a. Pain
    b. Urethral discharge
    c. Ecchymoses
    d. Priapism
    e. Preputial swelling
27. Which of the following features of Buschke-Löwenstein tumor characterizes it as different from condyloma acuminatum?
    a. Propensity for early distant metastasis
    b. Disruption of the rete pegs
    c. Loss of pigmentation
    d. Autoamputation
    e. Invasion and destruction of adjacent tissues by compression
28. All of the following statements are true regarding the management of bulky inguinal metastases in patients with squamous penile carcinoma EXCEPT:
    a. cross-sectional imaging plays an important role in establishing the extent of disease.
    b. neoadjuvant chemotherapy with a cisplatin-containing regimen is recommended.
    c. post-chemotherapy surgical consolidation to achieve disease-free status is recommended following an objective response to chemotherapy.

d. among patients who progress through chemotherapy, salvage surgery can often result in prolonged survival.
e. chemo-radiation for tumor cytoreduction prior to surgical resection or as definitive treatment may play a role.

29. Small lesions of erythroplasia of Queyrat may be successfully treated with which of the following?
    a. Topical 5% 5-FU
    b. Neodymium:yttrium-aluminum-garnet (Nd:YAG) laser
    c. Local excision
    d. Imiquimod
    e. All of the above

Answers

1. **a. Balanitis xerotica obliterans.** Also known as lichen sclerosus, manifests as a whitish patch on the prepuce or glans, often involving the meatus and sometimes extending into the fossa navicularis. The cause is unknown but has not been associated with known viruses. Condyloma, Bowenoid papulosis, and erythroplasia of Queyrat are associated with HPV infection while Kaposi sarcoma is associated with HPV 8 infection.
2. **d. Human papillomavirus (HPV) infection.** HPV is recognized as the principal etiologic agent in cervical dysplasia and cervical cancer.
3. **d. Location.** Carcinoma in situ of the penis is referred to by urologists and dermatologists as erythroplasia of Queyrat if it involves the glans penis or prepuce and as Bowen disease if it involves the remainder of the penile shaft skin, genitalia, or perineal region.
4. **b. Human herpesvirus (HHV) type 8.** HHV type 8—also known as Kaposi sarcoma-associated herpesvirus—is strongly suspected to be the etiologic agent of epidemic (AIDS-related) Kaposi sarcoma.
5. **a. Glans.** Penile tumors may present anywhere on the penis but occur most commonly on the glans (48%) and prepuce (21%).

6. **d. Gonorrhea.** No convincing evidence has been found linking penile cancer to other factors such as occupation, other venereal diseases (gonorrhea, syphilis, herpes), marijuana use, or alcohol intake.
7. **a. Circumcision after 21 years of age.** Adult circumcision appears to offer little or no protection from subsequent development of the disease. These data suggest that the crucial period of exposure to certain etiologic agents may have already occurred at puberty and certainly by adult age, rendering later circumcision relatively ineffective as a prophylactic tool for penile cancer.
8. **b. Because of the associated discomfort, patients usually present to physicians within the first month of noting the lesion. Patients with cancer of the penis, more than patients with other types of cancer, seem to delay seeking medical attention.** In large series, from 15% to 50% of patients have been noted to delay medical care for more than a year.
9. **d. Tumor stage, grade, and vascular invasion status all provide prognostically important information.** Confirmation of the diagnosis of carcinoma of the penis, an assessment of the structure(s) invaded, and tumor grade by the combination of an adequate biopsy, and complete clinical assessment are beneficial before the initiation of definitive therapy. Biopsy can be performed as a frozen section immediately before definitive therapy in some cases.
10. **e. Metastasis initially involves inguinal lymph nodes above the fascia lata.** The lymphatics of the prepuce form a connecting network that joins with the lymphatics from the skin of the shaft. These tributaries drain into the superficial inguinal nodes (the nodes external to the fascia lata).
11. **c. It may be due to the action of parathyroid hormone-like substances released from the tumor.** Parathyroid hormone and related substances may be produced by both tumor and metastases that activate osteoclastic bone resorption.
12. **a. Both ultrasonography and MRI lack sensitivity for the detection of corpus cavernosum involvement.** The sensitivity of ultrasonography for detecting cavernosum invasion was 100% in

one study. This study confirmed the value of ultrasonography in assessing the primary tumor also reported by other investigators. For lesions suspected of invading the corpus cavernosum, both ultrasonography and contrast-enhanced MRI may provide unique information, especially when organ-sparing surgery is considered.

13. **c. Stage T2 tumors invade the corpus spongiosum but not the cavernosum** According to the 8th edition, staging system invasion of the corpus cavernosum is now designated T3, indicating a worse prognosis than tumors involving the spongiosum alone.
14. **e. The extent of lymph node metastasis.** The presence and extent of metastasis to the inguinal region are the most important prognostic factors for survival in patients with squamous penile cancer.
15. **e. A single metastasis of only 6 cm.** Pathologic criteria associated with long-term survival after attempted curative surgical resection of inguinal metastases (i.e., >70% 5-year survival) include (1) minimal nodal disease (up to two involved nodes in most series), (2) unilateral involvement, (3) no evidence of extranodal extension of cancer, and (4) the absence of pelvic nodal metastases. A lymph node larger than 4 cm is often associated with extranodal extension of cancer.
16. **e. Stage Ta tumors.** Tumor histologic type associated with little or no risk for metastasis includes those patients with primary tumors exhibiting (1) carcinoma in situ or (2) verrucous carcinoma.
17. **c. Standard ilioinguinal dissection.** In patients with no evidence of palpable adenopathy who are selected to undergo inguinal procedures by virtue of adverse prognostic factors within the primary tumor, the goal is to define whether metastases exist with minimal morbidity for the patient. A variety of treatment options for this purpose have been reported and include (1) dynamic sentinel node biopsy, (2) superficial dissection, (3) modified complete dissections, and (4) laparoscopic and robotic approaches.
18. **e. b and c only.** Both superficial inguinal lymph node dissection and dynamic sentinel lymph node biopsy (latter at experienced centers) are considered standard procedures for determining the presence of inguinal microscopic metastases. Biopsies directed to a

specific anatomic area (i.e., sentinel lymph node biopsy) can be unreliable in identifying microscopic metastasis and are no longer recommended.

19. **b. A contralateral staging procedure is not indicated.** Support for a bilateral procedure is based on the finding of bilateral lymphatic drainage from the primary site in the majority of cases and contralateral metastases in more than 50% of patients so treated in some series, even if the contralateral nodal region was negative to palpation.
20. **d. Two unilateral inguinal nodes with focal metastases.** For patients requiring ilioinguinal lymphadenectomy because of the presence of metastases, adjuvant chemotherapy should be considered for those exhibiting more than two positive lymph nodes, extranodal extension of cancer, or pelvic nodal metastasis. Reports from one center further confirmed the value of adjuvant chemotherapy. Of 25 node-positive patients treated with adjuvant combination vincristine-bleomycin-methotrexate chemotherapy, 82% survived 5 years, compared with 37% of 31 patients treated with surgery alone.
21. **c. Squamous cell carcinoma.** The majority of tumors of the penis are squamous cell carcinomas demonstrating keratinization, epithelial pearl formation, and various degrees of mitotic activity.
22. **a. Bleomycin.** Response rates of bleomycin, whether as a single agent or in combination with other agents, has not been shown to be superior to cisplatin alone, but has been associated with significant pulmonary toxicity and death in several series of patients treated for metastatic penile cancer.
23. **e. a, b, and c.** Radiation therapy may be considered in a select group of patients: (1) young individuals presenting with small (2 to 4 cm), superficial, exophytic, noninvasive lesions on the glans or coronal sulcus; (2) patients who refuse surgery as an initial form of treatment; and (3) patients with distant or inoperable metastases who require local therapy and express a desire to retain the penis.
24. **a. Penile skin is protected from exposure to the sun. Melanoma and basal cell carcinoma rarely occur on the penis, presumably because the organ's skin is protected from exposure to the sun.**

25. **b. Diffuse disease.** When lymphomatous infiltration of the penis is diagnosed, a thorough search for systemic disease is necessary.
26. **d. Priapism.** The most frequent sign of penile metastasis is priapism; penile swelling, nodularity, and ulceration have also been reported.
27. **e. Invasion and destruction of adjacent tissues by compression.** The Buschke-Löwenstein tumor differs from condyloma acuminatum in that condylomata, regardless of size, always remain superficial and never invade adjacent tissue. **Buschke-Löwenstein tumor displaces, invades, and destroys adjacent structures by compression. Aside from this unrestrained local growth, it demonstrates no signs of malignant change on histologic examination and does not metastasize.**
28. **d. Among patients who progress through chemotherapy salvage surgery can often result in prolonged survival.** Surgery among patients who do not respond to chemotherapy is commonly associated with death due to either rapidly occurring locoregional recurrence or distant metastases
29 **e. All of the above.** When lesions are small and noninvasive, local excision, which spares penile anatomy and function, is satisfactory. Circumcision will adequately treat preputial lesions. Fulguration may be successful but often results in recurrences. Both Topical 5-FU as the 5% base and Imiquimod causes denudation of malignant and premalignant areas while preserving normal skin. There are also reports of successful treatment with Nd:YAG laser.

Chapter Review

1. HPV infection among men is associated with penile carcinoma, as well as with cervical dysplasia and carcinoma among female partners.
2. Imiquimod cream is now the standard topical treatment of choice for condyloma.
3. Giant condyloma of Buschke Lowenstein are locally progressive HPV related tumors that are nonmetastatic but require complete

excision and follow-up. The potential for coexisting or malignant degeneration to squamous carcinoma has been shown.

4. Penile carcinoma in situ (Tis) is an intraepithelial malignant process manifesting as three differing clinical entities including Bowenoid papulosis, Bowen disease, and erythroplasia of Queyrat.
5. Penile cancer is rare in developed countries and varies worldwide with age, circumcision, HPV exposure, and lifestyle/hygiene practices.
6. Recent epidemiologic data from the United States suggest a disparity in both incidence and outcome of penile cancer for Puerto Rican Hispanic men.
7. Risk factors for development of penile cancer include lack of neonatal circumcision, phimosis, HPV infection, exposure to tobacco products, penile lichen sclerosus, and potentially penile trauma.
8. Penile cancer represents a preventable disease in most cases via neonatal circumcision, HPV vaccination, and/or behavior modification.
9. Penile cancer often begins on the surface of the glans penis or in the preputial area, where it progressively enlarges.
10. Delay both in seeking medical attention and then in subsequent definitive biopsy is common.
11. Examination of both the penile primary tumor and the inguinal region is critical to treatment planning.
12. Metastasis occurs by embolization of tumor deposits from the penile tumor through penile lymphatics to the inguinal lymph nodes.
13. Distant metastases occur late in the history of the disease.
14. Adequate tumor biopsy is essential to diagnosis and treatment planning.
15. Squamous carcinoma histologic subtypes have recently been classified by into two major groups by their relationship to HPV and show distinct morphologic features and clinical behavior.
16. Pathologic description of anatomic structures invaded (i.e., stage), the grade, and the status of vascular and perineural invasion provide important information to assess the risk of metastasis.
17. Soft-tissue detail of penile tumors is best imaged by MRI.

18. Physical examination provides the most reliable staging information for small distal lesions.
19. Penile MRI especially performed in combination with artificial erection may provide unique staging information when physical examination findings are equivocal.
20. Physical examination of the inguinal region remains the clinical gold standard for evaluating the presence of metastasis in the nonobese patient.
21. CT or MRI can be useful in evaluating the inguinal region of obese patients and in those who have had prior inguinal surgery.
22. Among patients with proven inguinal metastases, CT scan of the abdomen and pelvis may help determine those patients with poor prognostic features for cure with surgery alone.
23. Positron emission tomography (PET)/CT may be useful among patients with clinically detected inguinal metastases to define the presence of pelvic or distant metastasis.
24. Both clinical and pathologic factors related to the presence and extent of lymph node involvement determine survival and should be recorded.
25. The current, eighth edition unified TNM staging system represents a consensus document that includes both clinical and pathologic descriptors that provide important prognostic information.
26. Patients with small lesions of low grade and stage (Tis, Ta, T1; grade 1 and grade 2) are the optimal candidates for organ preservation to maintain sexual quality of life.
27. The goals of organ preservation are to maintain glanular tissue for sensory purposes when possible and/or to maintain penile length when glans penis preservation is not possible.
28. Surgical modalities include limited excision strategies, Mohs surgery, and laser ablation.
29. Local recurrence rates overall after organ preservation are higher than with traditional amputation; however, when local recurrences are detected and treated, early survival does not appear be adversely affected.
30. Amputation remains the standard for large or deeply invasive lesions, to gain rapid tumor control.

31. The presence and extent of inguinal metastases determine survival in penile cancer.
32. Patients with persistent palpable inguinal adenopathy should undergo an ultrasound or CT guided inguinal biopsy, followed by management appropriate for the clinical scenario.
33. On the basis of the histologic features of the primary tumor, the risk of lymph node metastases can be assessed in patients with no palpable adenopathy based upon the TMN staging system. Dynamic sentinel node biopsy (DSNB), superficial inguinal lymph node dissection (ILND), or close follow-up can be recommended based upon TMN stage and other histologic features.
34. Factors associated with a high cure in surgically treated patients include no more than two inguinal metastases, unilateral involvement, no extranodal extension (ENE) of cancer, and the absence of pelvic metastases. Patients with higher volumes of disease should be considered for adjuvant or neoadjuvant therapy.
35. Morbidity of lymphadenectomy is decreasing in contemporary series.
36. Superficial ILND reliably determines the presence of microscopic inguinal metastases without the need for specialized facilities but can have significant morbidity.
37. Modified DSNB techniques to determine microscopic inguinal disease exhibit low morbidity, have been validated externally in higher-volume centers, and are now a recommended procedure in such centers.
38. Laparoscopic and robotic ILND obtains lymph node yields that are comparable to those of open techniques when used in selected patients and are an appropriate staging procedure in clinically lymph node negative patients. Additional studies with larger patient numbers and longer follow-up are required before routine adoption into clinical practice among patients with clinically positive inguinal lymph nodes.
39. Pelvic lymph node dissection (PLND) is now recommended when more than one inguinal lymph node exhibits metastasis or when ENE of cancer is present.
40. Radiation provides an effective penile-preserving approach for T1 to T2 squamous cell carcinomas less than 4 cm using either external-

beam radiotherapy or brachytherapy.

41. Since 20% of recurrences occur after 5 years, continued follow-up is required as salvage penectomy for persistent or recurrent disease may be curative.
42. The criteria for surgical staging of inguinal lymph nodes are the same whether patients undergo primary radiation or primary surgical management.
43. Unresectable lymph nodes may be rendered operable by neoadjuvant chemotherapy or chemo-radiation.
44. Integration of radiation, surgery, and chemotherapy in advanced disease is being investigated in a prospective international randomized trial (InPACT: EA 8134).
45. Palliative radiotherapy may be beneficial for metastatic disease.
46. Neoadjuvant chemotherapy with a cisplatin-containing regimen should be considered for patients with lymph node metastases, as responses in this setting may facilitate curative resection. In the absence of Level 1 evidence, the optimal or standard multimodal strategy remains undefined.
47. The use of bleomycin in the treatment of men with metastatic penile cancer was associated with an unacceptable level of toxicity and is discouraged.
48. Surgical consolidation to achieve disease-free status or palliation should be considered in fit patients with a proven objective response to systemic chemotherapy.
49. Among patients who progress through chemotherapy, surgery is not recommended.
50. Basal cell carcinoma represents a highly curable variant with a relatively low metastatic potential.
51. Sarcomas are prone to local recurrence; regional and distant metastases are rare. Superficial lesions can be treated with less radical procedures.
52. Melanoma is an aggressive form of cancer but can be cured if diagnosed and treated with the appropriate surgical procedure at an early stage. Novel immunotherapy strategies may improve survival in recurrent or advanced disease.

53. Extramammary Paget disease (EMPD) disseminates by intraepidermal spread initially. Wide local excision to achieve negative margins is the therapy of choice. Invasive EMPD can be lethal.
54. Penile metastases most often represent spread from a clinically obvious existing primary tumor. Prognosis is poor, and therapy should be directed toward the primary tumor site histology and local palliation.